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Maintaining E-commerce supply chain viability: Addressing supply risks with a strategic live-streaming channel☆

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ABSTRACT

Under the recent COVID-19 pandemic and operational disruption, we have witnessed the importance of maintaining supply chain viability. Exploring the channel effect (e.g., the use of innovative sales channels) in addressing supply risks is essential to understanding the viability of e-commerce supply chains. Motivated by the emergence of live-streaming channels in the "new normal", in this study, we consider an e-commerce supply chain facing supply risks. Capturing supply risks by a random supply viability, we investigate the channel decision on whether to introduce live streaming and the platform's optimal selling format selection (between reselling and marketplace). We prove that live streaming can help address supply risks because it can utilize the streamer's sales effort to impact the live-streaming retail price and create more flexibility. We confirm that a sufficiently high live-streaming efficiency is crucial for introducing a live-streaming channel. Interestingly, we uncover that high volatility of supply risks incentivizes the adoption of live streaming. We also find that incorporating live streaming may change the e-tailer's selling format preferences because the reselling format exhibits a higher acceptance of live streaming than the marketplace one, which is further influenced by the combined influence of live streaming and supply risks. Our results imply that live streaming plays a positive role in maintaining supply chain viability by enhancing the supply chain's overall ability to address supply risks. This paper provides scientific evidence on the importance of having a live-streaming channel in addressing supply risks in the new normal.

1. Introduction

1.1. Background and motivation

The COVID-19 pandemic and various operational disruptions have accelerated the development of live-streaming e-commerce [1–3]. China's live-streaming e-commerce market reached nearly five trillion yuan in 2023, a twelvefold rise from before the crisis. It is forecasted that the number of live-streaming users in China will be over 430 million by 2026. The U.S.'s livestream e-commerce sales reached 50 billion dollars in 2023 and are expected to grow by 36 percent over the next three years.[1] Live streaming has revolutionized e-commerce, with digital tools being applied to communicate with consumers more effectively. This helps resolve the consumer valuation uncertainty problem [4,5] and enhances the consumer shopping experience [6,7].

In the past five years, many e-commerce platforms have incorporated live streaming into their business operations and achieved huge success [8–10]. Amazon, the live-streaming selling of which was launched in 2019, engaged over 1 billion customers in the U.S. and India to shop with Amazon Live in 2023.[2] Tik Tok, a popular social media platform, officially launched Tik Tok Shop in the U.S. in 2023,[3] aiming to expand its global e-commerce business by leveraging the live-streaming

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channel.[4] Alibaba, the e-commerce giant in China and a pioneer of live-streaming selling, reported that live-streaming accounted for over 60 % of the channels with over RMB100 million in GMV during the most recent Singles' Day shopping festival in 2023.[5]

Live-streaming selling has profoundly revolutionized the e-commerce business mode. The e-commerce business mode commonly applies two different selling formats: (i) reselling, where the e-tailer purchases products from the brand and resells them to customers, and (ii) the marketplace format, where the brand sells directly to customers. A key distinction between these two formats lies in the allocation of control rights over noncontractible decisions, such as pricing, customer service, and order fulfillment [11–14]. A live-streaming channel can be introduced by either the e-tailer under the reselling format or the brand under the marketplace format. For example, JD's product managers effectively recommend their products through live streaming, leading to significant sales boosts.[6] Similarly, the apparel brand Zara engages in live streaming on e-commerce platforms, attracting a large audience and enhancing its sales reach.[7] Consequently, the growing popularity of live-streaming selling raises new considerations for the introduction of a live-streaming channel and the corresponding adjustments in the platform's selling format.

The outbreak of the COVID-19 pandemic has confirmed the importance of maintaining supply chain resilience and viability [15–19]. The unprecedented disruptions have severely impacted multiple aspects of supply chain management [20]. Government containment policies and the spread of infection have led to material and labor shortages, as well as delivery delays, leading to increasing and drastically rising supply risks [21–23]. Under the e-commerce framework, strategies like platforms' selling format selection [24], the production planning design [25], and supply chain structures integration [26] can be applied to maintain the supply chain viability. However, incorporating live-streaming selling as the new normal in the post-pandemic era brings a new approach to keeping viability. The live-streaming market power influences its selling effectiveness and efficiency [8–10]. Low consumer awareness and acceptance of live streaming may incentivize the abandonment of live-streaming selling.[8] Streamer's ability to sell and sales effort also determine the promotion outcome of live streaming [5,27,28]. Unqualified streamers may raise challenges to the live-streaming market.[9] Therefore, the integration of live-streaming selling raises new considerations for maintaining the viability of the e-commerce supply chain and, to the best of our knowledge, remains unclear in the literature.

In this paper, we explore the viability of an e-commerce supply chain with live-streaming selling by investigating its role in coping with supply risks. Both practice and literature evidence have confirmed the benefits of using live-streaming selling to approach demand. However, the benefits of introducing live-streaming selling rely on many factors, including the market acceptance of the channel, the live-streaming commission, the streamer's ability to sell, the input sales effort, and even the platform's selling format. We explore the necessity of introducing a live-streaming channel by addressing the factors mentioned above, and more importantly, we demonstrate the impact of supply risks to explore the viability challenge of e-commerce supply chains.

1.2. Research questions

Motivated by the importance of maintaining the viability of e-commerce supply chains and the popularity of live-streaming selling in practice, we construct a stylized e-commerce supply chain model with live-streaming selling and supply risks to investigate the following research questions (RQs).

RQ1: What is the role of live streaming in addressing supply risks? Will it help to maintain supply chain viability? Are there any crucial factors impacting its performance?

RQ2: How to decide whether to introduce a live-streaming channel or not? How do supply risks influence this decision?

RQ3: What role do the platform's selling formats (i.e., reselling and marketplace) play? Is there any difference between the two formats in incorporating live streaming and maintaining the supply chain viability?

To address the above research questions, we construct an analytical e-commerce supply chain model facing supply risks. We characterize supply risks by a random supply viability, representing the yield rate of the supply. We feature the steamer's ability to sell, endogenize the sales effort, and consider "segmented customers influenced differently" by live-streaming selling. We construct a two-period model where the retail prices and the streamer's sales effort are short-term decisions, and the live-streaming channel adoption and the platform's selling format selection are long-term strategies. Given the optimal short-term decisions, we study the conditions under which a live-streaming channel should be introduced. Then, we further analyze the influence of the platform's selling format and discuss the e-tailer's preference over it. The role of incorporating live streaming for the viability of the supply chain and the joint effect with the platform's selling format are discussed. We also check the robustness of the results by considering different live-streaming commission structures and examining the impact of viability improvement.

We find that live-streaming selling enhances the supply chain's ability to address supply risks by leveraging the streamer's sales effort to influence the demand market. In traditional channels, supply risks typically harm the performance of e-commerce participants. We prove that introducing a live-streaming channel can mitigate these negative effects of supply risks and achieve an improvement in e-commerce participants' performance if the expected supply viability is not too low. We confirm that introducing a live-streaming channel requires a reasonably high live-streaming efficiency, which decreases as the variance of supply viability increases. That is, the volatility of supply risks will incentivize the adoption of a live-streaming channel. We also find that a higher acceptance of a live-streaming channel is more likely to appear under the reselling format because the profit from the live-streaming channel is not deducted as part of the platform commission as it is under the marketplace format. Furthermore, the e-tailer's preference over the platform's selling format typically depends on the platform's commission rate. The varying acceptance of the two selling formats over the live-streaming channel complicates the e-tailer's selling format preference, which is further affected by the joint effect of the random supply viability.

1.3. Organization

The rest of the paper is organized as follows. A concise literature review is provided in Section 2. Section 3 introduces the model settings on live streaming, e-commerce selling format, and supply risks. In Section 4, we conduct the analysis, where we first derive the optimal short-term decisions and then analyze the long-term channel and format choices against the supply risks. In Section 5, we consider two extended models examining the impact of live-streaming commission structure and supply viability improvements. Section 6 concludes the paper. All proofs are relegated in the Appendix.

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2. Literature review

This paper relates to three research streams: supply chain viability and resilience, live-streaming e-commerce, and the platform's selling format.

2.1. Supply chain viability and resilience

The COVID-19 pandemic has triggered new research areas in supply chain resilience and viability [16,29]. Supply chain resilience refers to the ability to recover performance from short-term disruptions [22]. Supply chain viability is an extended concept to "maintain itself and survive in a changing environment over a long period" [17–19,30,31]. Various methods have been proposed to explore the viability of supply chains, including empirical approaches [20,32], optimization and simulation [33,34], as well as game-theoretic analytical models to analyze operational decisions under supply risks and evaluate supply chain viability [21,23].

In this paper, we consider the viability of e-commerce supply chains after disruptions such as the COVID-19 pandemic. Some recent empirical studies discuss the resilience of the e-commerce business. Guthrie et al. [35] employ a descriptive case study to explore how consumer resilience, including reacting to the pandemic, coping with new behavior, and adapting to new consumption habits, influences the evolution of online purchasing behavior during the COVID-19 pandemic. Han et al. [36] study the sales data from Alibaba during the pandemic and show that the recovery pattern can indicate the digital resilience of e-commerce business and the importance of logistics capacity during the COVID-19 pandemic. E-commerce supply chain viability is also investigated using a mathematical modeling framework. For example, Paul et al. [25] develop a recovery planning optimization model for online business operations affected by COVID-19 by considering dynamic and uncertain scenarios in a finite planning horizon. Xu et al. [24] investigate the selling formats (reselling and marketplace) under a cap-and-trade regulation with demand disruption. In this paper, we use a game-theoretical approach to examine the impact of supply viability on the live-streaming strategy and selling format choice, which complements the above-mentioned studies.

2.2. Live-streaming e-commerce

Live streaming has received great attention in recent years. It is usually associated with the celebrity [37]. It is a relatively new retailing channel, which is distinctly different from the traditional website and mobile apps e-commerce channels [38]. Many empirical studies (e.g., [6,39]) and analytical studies (e.g., [1,5,40]) have been conducted to capture live-streaming features and investigate operational decisions. For this stream of literature, we review those studies discussing whether or not it is wise to introduce live streaming under the e-commerce platform framework. For some discussion on evolution of live streaming in operations, we refer readers to Yang et al. [41].

The existing studies have considered whether to provide a live-streaming service to the existing selling channel [42]. Cui et al. [4] analytically explore a seller selling through a marketplace platform and discuss whether the live-streaming service should be introduced by characterizing consumers' hassle cost. They find that both the seller and the platform prefer adopting live streaming if the hassle cost is low. Niu et al. [2] consider a brand selling through a retailer and a self-operated direct channel, where a key opinion leader can be introduced to the latter. By evaluating the tradeoffs among the crucial opinion leader's network effect, order retain rate, and commission rate, the conditions for adopting the live-steam service are obtained. Yang et al. [27] develop a dual-channel supply chain where a manufacturer can sell directly through its website (direct channel) and also sell via an e-retailer (reselling channel). Either of the two channels may adopt the live-streaming service, and there exist spillover effects between the two channels. They reveal how the manufacturer's preferences over incorporating live streaming relate to consumers' channel preferences. Xie et al. [3] study the case where a retailer sells its store brand products as well as national brand products, where the former can be promoted with live streaming by the retailer, and the latter can be promoted with live streaming by either the retailer or the manufacturer. Different live-streaming promotion strategies towards the two products are derived.

Some studies investigate the necessity of adding a live-streaming channel to the current selling structure. For example, Du et al. [8] consider a manufacturer selling through an e-retailer and choosing whether to add a live streaming channel operated by itself, which can cause some customers switch from the e-tailer. The critical thresholds on the streamer's sales ability and the proportion of switching consumers are identified for operating the live-streaming channel. Wang and Zhang [43] also consider a manufacturer selling through the e-tailer and deciding whether to introduce another live-streaming through the e-tailer. The authors investigate the impact of the potential of misfit elimination from live streaming and consumers' hassle cost on the channel decision.

2.3. The platform's selling format

The third related stream of literature is the selection of e-commerce platform selling formats, i.e., the selection among reselling, marketplace (agency), and hybrid, under different settings. Hagiu and Wright [12] are among the first to differentiate the three selling formats based on the allocation of control rights and to explore the impact of marketing activity on the selling format choice. Many following studies then discuss the format strategies under different settings.

Some consider e-commerce platforms' format decisions. Abhishek et al. [11] consider a manufacturer selling through a traditional brick-and-mortar retailing channel and two e-tailers and show that the e-tailers' selection between reselling and agency selling depends on the demand spillover effect between the two channels. Kwark et al. [44] investigate an online retailer's format choice for two substitutable products by considering the product information on the quality and fit dimensions. It is shown that both the level of the precision of the information and the dominance between the quality dimension and the fit dimension will affect the online retailer's preference. Tian et al. [14] further investigate the case where an online intermediary can provide different selling modes to two competing suppliers with the consideration of order-fulfillment costs. They show that the marketplace mode is optimal for low order-fulfillment costs with light competition, the reseller mode better fits high order-fulfillment costs under intense competition, and the hybrid for the middle. The selling format may also be the users' decision, which normally incorporates the platform's advantages regarding demand information. For example, Tsunoda and Zennyo [45] show that the platform's demand for information sharing incentivizes the supplier's adoption of agency selling, and Ha et al. [46] show that the e-tailer's information sharing and the manufacturer's encroachment on the market are complementary. Besides the above-mentioned studies, the selling format choices are also investigated with returns [47,48], logistics services [49], service efficiency [50], the advanced technologies [51,52], etc.

A few papers explore selling format strategies with the consideration of live-streaming selling, as in the case discussed in our paper. For instance, Ha et al. [53] consider a manufacturer selling through an online intermediary, where the intermediary can invest in live-streaming retailing. They explore the impact of platform commission rate and the retailing investment cost on the intermediary and the manufacturer's preference over reselling, marketplace, and hybrid formats. The authors prove that although the intermediary and the manufacturer may have diversified preferences over the three formats, for overall performance, the hybrid format can be more efficient than the other two.

The most related studies to our paper are Hao and Yang [9] and Ji

et al. [54], both of which consider the platform's format choice between reselling and marketplace, where the live-streaming service is operated by the platform (brand) under the reselling (marketplace) format in addition to a traditional online channel without live streaming. To be specific, Hao and Yang [9], given the existence of the live-streaming channel, consider three different pricing strategies for the traditional online and the live-streaming channel, namely the exact high, the same low, and the differential strategy. The optimal pricing strategies under different formats are discussed, further supporting the format choice. Ji et al. [54], taking the introduction of a live-streaming channel as an endogenous decision as in our model, further discuss the price discount strategy for the live-streaming channel. In particular, they consider the possibility of regular price (i.e., the price for the traditional online channel) to change with the adoption of the live-streaming channel. The regular price can be committed to being stable (referred to as the committed regular price scheme) where only the discount rate to the live-streaming price is decided, and the regular price can also be dynamic (referred to as the dynamic price scheme) with the adoption of the live-streaming channel where the regular price and the discount rate are both selected. They show that the dynamic price scheme can promote live-streaming selling, and adopting live-streaming selling can incentivize the selection of marketplace format.

This paper also considers the live-streaming channel strategy and selling format preference under an e-commerce framework. Unlike the above-reviewed papers, we conducted the research in the presence of supply risks. We examine whether to introduce a live-streaming channel under (i) the supply risks with random supply viability and (ii) different platform selling formats. Novel insights are hence derived.

Contribution statement: To the best of our knowledge, ours is the first paper to investigate the role of live streaming in addressing supply risks and maintaining e-commerce supply chain viability. Our results imply that live streaming plays a crucial and positive role in maintaining supply chain viability by enhancing the supply chain's overall ability to address supply risks. Important insights with managerial implications are also derived.

3. Model

3.1. Live-streaming e-commerce

Live-streaming selling has been developed rapidly and is widely adopted by many major e-commerce retailers. In this paper, we consider a typical e-commerce supply chain with a brand and an e-tailer. The brand sells through the e-tailer under two possible selling formats. The first is the reselling format, where the e-tailer purchases products from the brand and resells them to the customers. The second is the marketplace format, under which the e-tailer operates as an agent, the brand selects the retail price and sells to the customers directly, and the e-tailer charges a proportional commission fee regarding the brand's revenue, referred to as the platform commission fee. Besides the traditional e-commerce channel (referred to as traditional channel for short hereafter), the seller (either the e-tailer or the brand), who holds higher visibility and influence, can introduce a live-streaming channel and hire a streamer to promote products in detail. We assume that the live-streaming channel is exerted by (i) the e-tailer under the reselling format [3,53], where the e-tailer controls pricing and compensates the streamer, and (ii) the brand under the marketplace format [4,43], where the brand retains control over pricing and pays the streamer. This distinction reflects the typical practice in e-commerce, where the control rights are allocated differently depending on the chosen selling format [12]. See Fig. 1 for the illustration.

To establish a live-streaming channel, an influential streamer is hired to conduct live video introductions and interact with the customers to promote the products. The streamer uses her/his experience, expert suggestions, and social influence to promote the products in detail, which helps resolve customers' valuation uncertainty for online products [55] and enhance the shopping experience [39,40]. Thus, customers gain extra utility from live-streaming shopping [5,6,56]. The additional value from the live streaming relates to the streamer's sales ability and the streamer's sales effort. The higher the streamer's ability, the higher the quality of the promotion activities. The greater the streamer's sales effort, the better the shopping experience the customers gain. Let θ be the streamer's sales ability, and let e be the streamer's sales effort. We consider that live-streaming promotion will increase customers' valuation by θe and incur a cost of $\frac{1}{2}e^2$ [2,27,53].

To compensate for the streamer's effort, the e-tailer/brand needs to pay the streamer a live-streaming commission fee. There are several commonly adopted contractual forms, like a lump-sum commission [57,56], proportional commission [2,27,40], and unit commission fee [54,58,59]. In the main model, we adopt the proportional commission scheme, where the streamer is paid a proportion $\alpha \in (0,1)$ of the revenue from the live-streaming channel. This proportional α is also regarded as the live-streaming commission rate. Other commission schemes will be discussed in the extended model for robustness check.

3.2. Customers

In practice, live streaming may have different impacts on different customers, which raises concerns about the potential negative effects associated with the introduction of a live-streaming channel. While some customers actively engage with the live-streaming channel, others may choose not to participate due to the time required to watch or a lack of awareness of the live-streaming channel [9,10,43]. To capture this feature, we assume that among the total market size, which is assumed to be 1. A proportion δ of customers, where $\delta \in (0,1)$, are willing to participate in live-streaming selling with negligible hassle costs, while the rest $1-\delta$ proportion of customers face extremely high participation costs and thus only have access to the traditional channel. Note that the

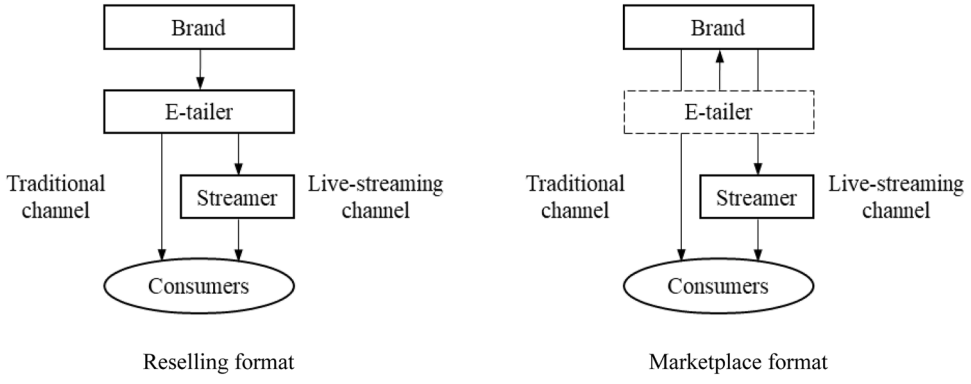


Fig. 1. Channel structure under different selling formats.

proportion δ can also be regarded as the market impact of the live-streaming channel, representing the proportion of customers who can reach the live streaming. Note that the proportion δ can also be regarded as the market impact of the live-streaming channel, representing the proportion of customers who can reach the live streaming.

We assume customers are heterogenous in their purchasing utility, denoted as v. It is assumed to be uniformly distributed over $[0,1]$, i.e., $v \sim U[0,1]$. Customers will strategically choose to purchase from the traditional channel, the live-streaming channel, or neither, depending on their purchasing utilities. Let p_t (p_l) be the retail price of the traditional (live-streaming) channel, and u_t (u_l) be the corresponding utility of purchasing. Then, we have $u_t = v - p_t$ and $u_l = v + \theta e - p_l$. To ensure that at least some customers choose to purchase from the two selling channels, we only consider those $p_t < 1$ and $p_l - \theta e < 1$. Furthermore, we restrict our attention to the case where $p_l - \theta e \le p_t$ to avoid the trivial case where all the customers are aware of the live-streaming channel and will also choose the traditional channel so that all the customers purchase from the traditional channel [10]. Recall that among the potential customers with size 1, a proportion of δ will consider the live-streaming channel as their option and the other proportion of $1-\delta$ will only consider the traditional channel. Thus, among all the potential customers, a fraction of $(1-\delta)(1-p_t)$ will choose the traditional channel and a fraction of $\delta(1-p_l+\theta e)$ will select the live-streaming channel.

Impacted by emerging events like the COVID-19 pandemic, the supply chain faces severe supply risks. To characterize supply risks, we let z be the supply viability, where z is a random variable with domain $[0, 1]$, with mean $\mu > 0$, and variance σ^2. For any order quantity q, we assume the output is zq, and the received orders with be allocated to customers equally. Recall the consumer utility analysis for the traditional and the live-streaming channels. The realized demands for the two channels are denoted as d_t and d_l, are $d_t = z(1-\delta)(1-p_t)$ and $d_l = z\delta(1 - p_l + \theta e)$. Note that the supply viability parameter z can also be regarded as the random yield [60–62].

3.3. Timeline of the events

The e-tailer's selling format choice and the channel decision on introducing a live-streaming channel are operational strategies, which are made before the selling season with supply risks. To characterize this feature, we consider the timeline of the events illustrated in Fig. 2.

Long-term decisions: The e-tailer first chooses the selling format, either reselling or marketplace. Then, either the e-tailer or the brand decides whether or not to introduce a live-streaming channel in addition to the traditional channel.

Short-term decisions: After supply risks are realized, the e-tailer and the brand make short-term pricing decisions. Under the reselling format, the brand selects the wholesale price w, and then the e-tailer sets the retail price(s); and under the marketplace format, the brand determines the retail price(s) and pays the e-tailer at an exogenously given commission rate λ. Here, λ follows the industrial norm and relates to various factors such as product category, type of seller, etc. If a live-streaming channel is introduced, the retail prices include both the live-streaming price and the retail price for the traditional channel. With the live-streaming channel, the streamer chooses the sales effort e. The demands are finally realized.

As illustrated in Fig. 2, the short-term decisions come after the realization of supply risks. Let π_B^{XY}, π_E^{XY}, and π_S^{XY} be the short-term profits for the brand (B), the e-tailer (E), and the streamer (S), respectively, given that the selling format is X and the channel strategy is Y. Here, the selling format can be reselling (R) or marketplace (M), and the channel strategy can be either the traditional channel only (T) or "introducing a live-streaming channel in addition to the traditional channel" (L).

Under the reselling format, the e-tailer purchases from the brand, determines the retail prices, and hires the streamer to conduct live-streaming promotion if the channel is introduced. In this case, we have

$$\pi_B^{RL}(w) = w(d_t + d_l), \tag{1}$$

$$\pi_E^{RL}(p_t, p_l) = (p_t - w)d_t + (1-\alpha)(p_l - w)d_l, \tag{2}$$

$$\pi_S^{RL}(e) = \alpha(p_l - w)d_l - \frac{1}{2}e^2. \tag{3}$$

Under the marketplace format, the brand operates the traditional and the live-streaming channels through the e-tailer and pays the e-tailer at a platform commission rate λ, where $\lambda \in (0,1)$. In this case, we have

$$\pi_B^{ML}(p_t, p_l) = (1-\lambda)(p_t d_t + (1-\alpha)p_l d_l), \tag{4}$$

$$\pi_S^{ML}(e) = \alpha(1-\lambda)p_l d_l - \frac{1}{2}e^2. \tag{5}$$

Note that the streamer shares the proportion α of the brand's remaining revenue $(1-\lambda)p_l d_l$ from the live-streaming selling after reducing the commission fee to the e-tailer, and the e-tailer simply receives the commission fee without making any decisions. In practice, the commission rate α is typically set between 20 % and 30 %.[10] A high commission rate can significantly reduce the profits that brands earn from live streaming, resulting in losses that outweigh potential benefits. To ensure positive demand and avoid trivial cases, we assume that $\alpha < \frac{2}{3\delta\theta^2}$. This assumption is consistent with common business practices and is supported by many prior studies on live-streaming selling, such as those by Ji et al. [63] and Yang et al. [27].

4. Analysis

In this section, we first discuss the streamer's sales effort and the retail prices for the short-term optimal decisions. Then, we analyze the long-term decisions on whether or not to provide a live-streaming channel and the selection between reselling and marketplace formats. We explore four scenarios with two possible selling channels (traditional channel only and traditional channel together with the live-streaming channel) and two possible selling formats (reselling and marketplace). For simplicity, we use the superscript "XY" to label the four scenarios, where $X = R, M$ stands for the reselling format and marketplace format and $Y = T, L$ stands for the traditional channel only and the traditional channel together with the live-streaming channel.

4.1. The optimal short-term decisions

As illustrated in Fig. 2, the short-term decisions are determined after the realization of supply risks, including the retail prices and the streamer's sales effort. We consider the reselling format first, followed by the marketplace format. As demonstrated in Section 3, there are four different scenarios XY, where $X = R, M$ and $Y = T, L$. The two cases with the purely traditional channel are standard, and we skip them to save space. In the following, we present the optimal short-term decisions to the two scenarios with both traditional and live-streaming channels.

Consider the reselling format with a live-streaming channel (i.e., the scenario RL). The e-tailer purchases from the brand and hires the streamer to conduct live-streaming promotions. The brand determines the wholesale price w, and then the e-tailer decides the retail prices p_t and p_l. Given the retail prices and the live-streaming commission rate α, the streamer selects the sales effort e. According to the profit functions given by (1)-(3) and applying backwards induction, we first derive the optimal sales effort by maximizing the streamer's profit $\pi_S^{RL}(e)$, then we analyze the e-tailer's profit $\pi_E^{RL}(p_t, p_l)$ to determine the retail prices. Finally, we maximize the brand's profit $\pi_B^{RL}(w)$ to derive the optimal

[10] https://daxueconsulting.com/international-brands-use-live-stream-shopping-to-advertise-in-china/ (accessed 30 October 2024)

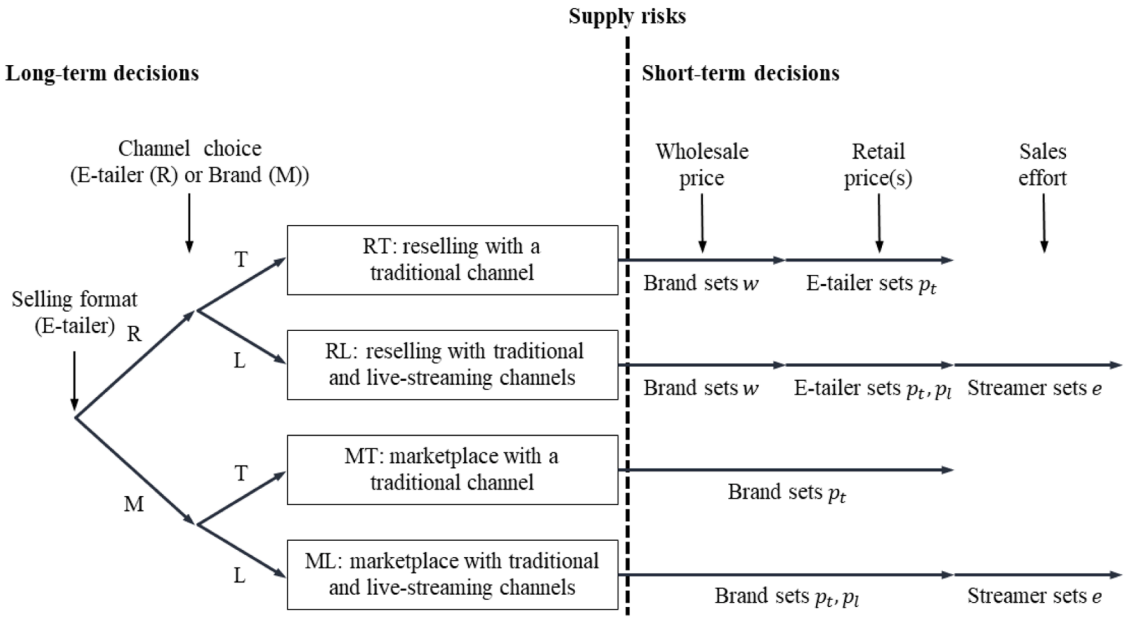


Fig. 2. Timelines of the events.

wholesale price. We summarize the results in the following lemma.

Lemma 1. *Under the reselling format with the live-streaming channel, the streamer's optimal sales effort is* $e^{RL} = \frac{z\alpha\delta\theta}{4(1-z\alpha\delta\theta^2)}$, *the optimal retail prices are* $p_t^{RL} = \frac{3}{4}$ *and* $p_l^{RL} = \frac{3-2z\alpha\delta\theta^2}{4(1-z\alpha\delta\theta^2)}$ *and the wholesale price is* $w^{RL} = \frac{1}{2}$.

Lemma 1 indicates that the optimal short-term decisions for scenario RL, where the retail price for the traditional channel and the wholesale price are both independent of the live-streaming channel.

Next, we consider the marketplace format with a live-streaming channel (i.e., scenario ML). The brand operates the traditional and the live-streaming channels through the e-tailer, and the brand pays the e-tailer a proportion of its revenue by a platform commission rate λ. In this case, the brand chooses the selling prices p_t and p_l, and the streamer determines the sales effort e. Similarly, by applying backwards induction, we derive the optimal decisions under this scenario in the following lemma.

Lemma 2. *Under the marketplace format with the live-streaming channel, the streamer's optimal sales effort is* $e^{ML} = \frac{z\alpha\delta\theta(1-\lambda)}{2(1-(1-\lambda)z\alpha\delta\theta^2)}$, *and the optimal retail prices are* $p_t^{ML} = \frac{1}{2}$ *and* $p_l^{ML} = \frac{1}{2(1-(1-\lambda)z\alpha\delta\theta^2)}$.

From Lemmas 1 and 2, we can easily compare the optimal short-term decisions under the reselling format and marketplace format. We find that there are some similarities (see Corollary 1) as well as some differences (see Corollary 2).

Corollary 1. *Under both the reselling format and marketplace format,* $p_l^{XL} > p_t^{XL}$, *and* $D^{XL} = D^{XT}$, *for* $X = R, M$.

Regardless of the selling format, the optimal live-streaming price is greater than that of the traditional channel. Live-streaming promotion increases the consumer utility by enhancing the shopping experience but may also bring some hassle cost of reaching the service. In this study, we use two types of customers to characterize their difference in the hassle cost for reaching live-streaming channel, i.e., a proportion of $1-\delta$ customers incurring extremely high hassling costs and being loyal to the traditional channel and the rest δ proportion of customers incurring negligible hassling cost and regarding the live-streaming channel as another possible choice. In this way, the consumers choosing the live-streaming channel can always enjoy the extra utility from the interactions with the streamer, and thus the live-streaming price can be higher than the retail price of the traditional channel. Similar results are obtained, for example, Cui et al. [4] and Pan et al. [5].

Corollary 1 also indicates that introducing a live-streaming channel does not increase the total realized demand, and it only causes some customers to switch from the traditional channel to the live-streaming channel even if the price is higher. This shift occurs because certain high-value customers may still choose to purchase through the live-streaming channel, but the increase in sales from the new live-streaming channel is offset by the decline in the traditional channel. Therefore, the introduction of live streaming does not increase total realized demand but only leads to a reallocation of existing demand among various sales channels. A similar result is obtained by Pan et al. [5]. As we will show in the extension, the validity of this result depends on the contractual forms of the live-streaming commission fee. As the live-streaming commission scheme changes from a proportional commission fee to a unit commission fee, the overall demand may increase with a live-streaming channel.

Besides the common properties in Corollary 1, the difference between introducing a live-streaming channel under reselling and marketplace formats can also be identified from Lemmas 1 and 2. As we will show later, the term "$\delta\theta^2$" plays an important role in establishing the key analytical results. Recall that δ is the fraction of the potential customers aware of the live-streaming channel, and θ is the streamer's ability to sell. A higher δ indicates a greater reach of the live-streaming channel, and a higher θ indicates that the streamer can provide a greater extra value. The two variables capture both the size of the potential customer base and the effectiveness of the streamer in converting that customer base into actual sales, thereby allowing us to evaluate how efficient the live-streaming channel is in promotion. For simplicity, we define $\Gamma := \delta\theta^2$ hereafter, and refer it as the *live-streaming efficiency*. The squaring of θ highlights that improvements in the streamer's sales ability have a multiplicative effect on sales; a more skilled streamer is more effective at engaging and converting viewers, which captures real practices.

Corollary 2. *Comparing the optimal decisions under the reselling and marketplace formats, (i)* $p_t^{ML} < p_t^{RL}$, *and* $D^{ML} > D^{RL}$, *(ii) there exists a threshold* Γ_0 *on the live-streaming efficiency such that* $p_l^{ML} > p_l^{RL}$ *and* $e^{ML} > e^{RL}$ *if* $0 < \lambda < \frac{1}{9}$ *and* $\Gamma > \Gamma_0$.

To understand the meaning of Corollary 2, we first note that in the literature, we know that the marketplace format, where the brand sells directly to the customers by paying the platform commission fee, can mitigate the double marginalization effect resulting from the reselling format. In other words, the marketplace format yields lower retail prices and higher realized demands than the reselling format (see, for example, [11,14,45]). Corollary 2(i) implies similar results about the retail price of the traditional channel and the realized demand. However, surprisingly, Corollary 2(ii) shows the live-streaming channel may not enjoy this premium from the marketplace format; that is, the live-streaming retail price and the streamer's sales effort can be higher under the marketplace format if the platform commission rate is sufficiently low and the live-streaming efficiency is considerably high. In this case, the high live-streaming efficiency implies a high power of the live-streaming channel, and the low platform commission rate indicates a low impact on the marketplace format. Thus, the brand operates the live-streaming channel much closer to a reselling format. The deviation from the mitigation on the double marginalization effect under the marketplace format is also observed in Hao and Yang [9], where the return issue is considered in the live-streaming channel.

4.2. The channel choice

The optimal short-term decisions are analyzed in Section 4.1, based on which we proceed to study the long-term channel: whether or not to introduce a live-streaming channel. From the long-term perspective, see Fig. 2, the expectation of the supply viability needs to be considered. We let $\Pi_H^{XY} = E_z\left[\pi_H^{XY}\right]$ be the expected long-run profits for $H = E, B$ under the selling format $X = R, M$ and the channel choice $Y = T, L$. For notational convenience, we also let $\overline{\Pi}_H^{XY}$ be the long-run profits with perfect supply viability, i.e., the special case of π_H^{XY} by setting $z = 1$.

Given the complexity of the short-term profit functions, we approach the expected profits by using Taylor approximations, which have been rather widely adopted for decades (see [64]). It has been shown that second-order Taylor expansions can serve as useful approximations to expected utility; see, for example, Pulley [65] and Kroll et al. [66]. Take the expected profit $\Pi_E^{RL} = E_z\left[\pi_E^{RL}(z)\right]$ as an example. Recall that $\pi_E^{RL}(z)$ is given by

$$\pi_E^{RL}(z) = \frac{z\left(1-\alpha\left(\delta+z(1-\delta)\delta\theta^2\right)\right)}{16\left(1-z\alpha\delta\theta^2\right)} = \frac{z(1-\delta)}{16} + \frac{(1-\alpha)}{16\alpha\theta^2\left(1-z\alpha\delta\theta^2\right)} - \frac{(1-\alpha)}{16\alpha\theta^2}.$$

By using the first two terms of the Taylor-series expansion of $\pi_E^{RL}(z)$ about its mean μ to approximate the expectation, we have

$$\begin{aligned}\Pi_E^{RL} &= E_z\left[\pi_E^{RL}(z)\right] = E\left[\pi_E^{RL}(\mu)\right] + E\left[(z-\mu)\frac{\partial \pi_E^{RL}(\mu)}{\partial z}\right] + E\left[\frac{(z-\mu)^2}{2}\frac{\partial^2 \pi_E^{RL}(\mu)}{\partial z^2}\right] \\ &= \frac{\mu(1-\delta)}{16} + \frac{(1-\alpha)\mu\delta}{16\left(1-\mu\alpha\delta\theta^2\right)} + \left(\frac{(1-\delta)}{16} + \frac{(1-\alpha)\delta}{16\left(1-\mu\alpha\delta\theta^2\right)^2}\right)E[(z-\mu)] + \frac{(1-\alpha)\alpha\delta^2\theta^2}{16\left(1-\mu\alpha\delta\theta^2\right)^3}E\left[(z-\mu)^2\right] \\ &= \frac{\mu(1-\delta)}{16} + \frac{(1-\alpha)\delta\mu}{16\left(1-\alpha\delta\theta^2\mu\right)} + \frac{(1-\alpha)\alpha\delta^2\theta^2\sigma^2}{16\left(1-\alpha\delta\theta^2\mu\right)^3},\end{aligned} \tag{6}$$

where the last equation holds from $\mathrm{E}[(z-\mu)] = 0$ and $E\left[(z-\mu)^2\right] = \sigma^2$. Similarly, we can obtain

$$\Pi_B^{RL} = \frac{\mu}{8}, \tag{7}$$

$$\Pi_E^{ML} = \frac{\mu(1-\delta)\lambda}{4} + \frac{\lambda\delta\mu}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)} + \frac{\lambda(1-\lambda)\alpha\theta^2\delta^2\sigma^2}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)^3}, \tag{8}$$

$$\Pi_B^{ML} = \frac{\mu(1-\delta)(1-\lambda)}{4} + \frac{(1-\alpha)\delta(1-\lambda)\mu}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)} + \frac{(1-\alpha)(1-\lambda)^2\alpha\theta^2\delta^2\sigma^2}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)^3}. \tag{9}$$

Before proceeding to discuss the channel decision, we first explore the impact of supply risks with the application of live streaming by using the expected profit expressions given by (6)-(9).

Lemma 3. *(i) Without introducing a live-streaming channel, supply risks always harm the e-tailer and the brand's profits,* i.e., $\Pi_H^{XT} < \overline{\Pi}_H^{XT}$, $X = R, M$ *and* $H = E, B$. *(ii) With introducing a live-streaming channel, there exists a threshold* μ^R (μ^M), *which decreases in* σ^2, *such that supply risks harm the e-tailer's (brand's) profit if and only if the mean of the supply viability is lower than this threshold,* i.e., $\Pi_E^{RL} < \overline{\Pi}_E^{RL}$ $\left(\Pi_B^{ML} < \overline{\Pi}_B^{ML}\right)$ *if and only if* $\mu < \mu^R$ ($\mu < \mu^M$).

Lemma 3 implies that when only the traditional channel exists, supply risks always harm the e-tailer and the brand. In contrast, if a live-streaming channel is introduced, it can help hedge the supply risks if the expected supply viability is not too low. This is because, with the live-streaming channel, the live-streaming retail price can adjust the streamer's sales effort and further impact consumers' purchasing utility. In this way, the live-streaming retail price enjoys more flexibility than the retail price in the traditional channel. Specifically, if the supply viability decreases, the live-streaming price can be lowered, leading to a decrease in the sales effort but still retaining the critical point about consumers' purchasing utility. If the expected supply viability is not too low, then the live-streaming channel's adjustment power is enough to create better performance than without supply risks. Introducing a live-streaming channel improves the supply chain's overall ability to address supply risks. Furthermore, the thresholds on the expected supply viability decrease in the variance σ^2. This indicates that if the supply viability has a higher volatility, the live-streaming channel plays a more important role.

Next, we compare the expected profits with and without the live-streaming channel to discuss whether or not to introduce a live-streaming channel. Specifically, under the reselling (marketplace) format, comparing Π_E^{RT} and Π_E^{RL} (Π_B^{MT} and Π_B^{ML}) derives the optimal channel decision. For ease of presentation, we assume that introducing a live-streaming channel is preferred if there is no difference between introducing and not introducing. We obtain the following proposition.

Proposition 1. *(i) Under the reselling format, there exists a threshold* Γ^R *such that* $\Pi_E^{RL} \geq \Pi_E^{RT}$ *if and only if* $\Gamma \geq \Gamma^R$. *(ii) Under the marketplace format, there exists a threshold* Γ^M *such that* $\Pi_B^{ML} \geq \Pi_B^{MT}$ *if and only if* $\Gamma \geq \Gamma^M$, *where* $\Gamma^R < \Gamma^M$.

Proposition 1 shows that under each selling format, there exists a supply viability related threshold on the live-streaming efficiency. If the live-streaming efficiency is higher than the threshold, then a live-streaming channel should be introduced. Otherwise, simply keeping

the traditional channel is more profitable.

Comparing the thresholds on the live-streaming efficiency under the two selling formats, it is easy to see that, regardless of the selling format, a live-streaming channel should not be introduced if the live-streaming efficiency is too low (i.e., $\Gamma < \Gamma^R$) and should be introduced if the live-streaming efficiency is considerably high (i.e., $\Gamma \geq \Gamma^M$). However, the selling format matters if the live-streaming efficiency is moderate (i.e., $\Gamma^R \leq \Gamma < \Gamma^M$). In this case, introducing a live-streaming channel is beneficial under the reselling format but is not under the marketplace format. This is because, under the latter, the brand needs to share a proportional commission fee at rate λ with the e-tailer. This reduces the net benefit from introducing a live-streaming channel, and increases the least live-streaming efficiency required to set up a live-streaming channel. In this sense, the reselling format is more robust in introducing a live-streaming channel than the marketplace format.

Sensitivity analysis regarding the thresholds is conducted and the neat findings are summarized in the following proposition.

Proposition 2. *(i) Γ^M increases in λ; (ii) Γ^R and Γ^M both decrease in σ^2.*

Proposition 2(i) shows that under the marketplace format, the lowest live-streaming efficiency required for introducing a live-streaming channel increases in the platform commission rate λ. This is because a higher platform commission rate implies a lower net benefit for the brand from the live streaming, as a higher commission will be transferred to the e-tailer. In this case, it becomes more difficult to incentivize the brand to introduce a live-streaming channel, such that a higher live-streaming efficiency is required. Proposition 2(ii) shows that the thresholds decrease with the variance of the supply viability, which means that a higher volatility of the supply viability will incentivize the introducing of a live-streaming channel. This is consistent with the previous analysis that a live-streaming channel can enhance the supply chain's overall ability to address supply risks.

4.3. Selling format strategy

In this section, we discuss the selling format strategy. Given the analytical complexity, we first analyze the case with perfect supply viability and then discuss the impact of supply risks. When the two formats generate equal profit for the e-tailer, we will let reselling be the preferred choice for ease of presentation.

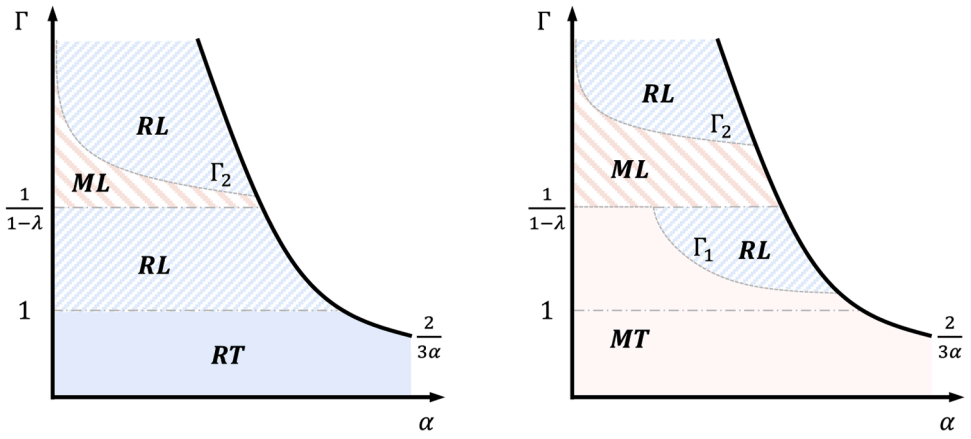
Without supply risks, the thresholds on the live-streaming efficiency in Proposition 1 can be simplified as $\Gamma^R = 1$ and $\Gamma^M = \frac{1}{1-\lambda}$. By comparing the e-tailer's profit under the two selling formats, we can fully derive the e-tailer's selling preferences as described in the following proposition.

Proposition 3. *Without supply risks, there exist two thresholds Γ_1 and Γ_2, where $\Gamma_1 < \Gamma_2$, such that (i) when $0 < \lambda \leq \frac{1}{4}$, the e-tailer prefers the marketplace format if $\frac{1}{1-\lambda} \leq \Gamma < \Gamma_2$, and (ii) when $\frac{1}{4} < \lambda < 1$, the e-tailer prefers the marketplace format if $\Gamma < \Gamma_1$ or $\frac{1}{1-\lambda} \leq \Gamma < \Gamma_2$.*

To visualize the meanings and findings of Proposition 3, we depict the e-tailer's preference over the selling format together with the channel decision in Fig. 3, which has two subcases depending on the platform commission rate λ. First, note that with the traditional channel only, it is easy to verify that the e-tailer prefers the reselling format if $\lambda \leq \frac{1}{4}$ and the marketplace format if $\lambda > \frac{1}{4}$, which remains true if the live-streaming efficiency is sufficiently low but becomes invalid as the live-streaming efficiency increases. If the platform commission rate is low (i.e., $\lambda \leq \frac{1}{4}$), see Fig. 3(a), the e-tailer prefers reselling only if the live-streaming efficiency is low to moderate. As the live-streaming efficiency increases to a moderate level, say $\frac{1}{1-\lambda} \leq \Gamma < \Gamma_2$, then the e-tailer will switch to the marketplace format until the live-streaming efficiency reaches a sufficiently high level. If the platform commission rate is high (i.e., $\lambda > \frac{1}{4}$), the marketplace format is preferred if the live-streaming efficiency is sufficiently low ($0 < \Gamma < \Gamma_1$), where a live-streaming channel will not be introduced under the marketplace format and the e-tailer benefits from the high platform commission fee. If the live-streaming efficiency increases and reaches Γ_1, the live-streaming efficiency is high enough so that introducing a live-streaming channel becomes the optimal channel strategy. This leads to a switch from marketplace format to reselling format because the e-tailer needs to use the reselling format to ensure the adoption of a live-streaming channel. When the live-streaming efficiency further increases to above the critical level $\frac{1}{1-\lambda}$, the adoption of a live-streaming channel is supported under the marketplace format, and thus the e-tailer switches back to the marketplace format to earn a high platform commission. Furthermore, if the live-streaming efficiency reaches a sufficiently high level, say $\Gamma \geq \Gamma_2$, the possible benefit from the live-streaming channel is significant such that the e-tailer prefers reselling so as to avoid the deduction of the platform commission fee.

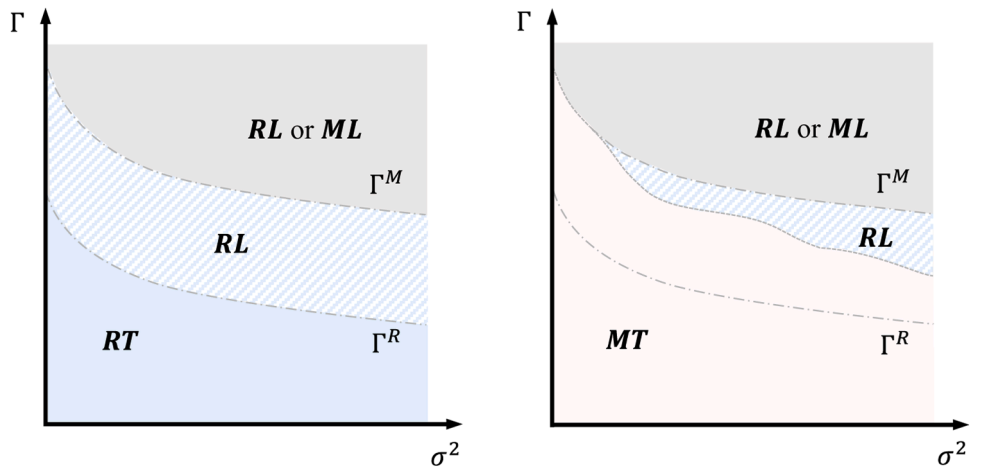
Next, we consider the impact of supply risks over the selling format. Given the corresponding optimal channel decision, comparing the e-tailer's profits under the two selling formats, we can derive the supply viability related thresholds on the live-streaming efficiency for the format strategy. The findings are shown in Proposition 4.

Proposition 4. *(i) When $0 < \lambda \leq \frac{1}{4}$, the e-tailer prefers the reselling format if $\Gamma < \Gamma^M$; (ii) when $\frac{1}{4} < \lambda < 1$, the e-tailer prefers the marketplace format if $\Gamma < \Gamma^R$.*



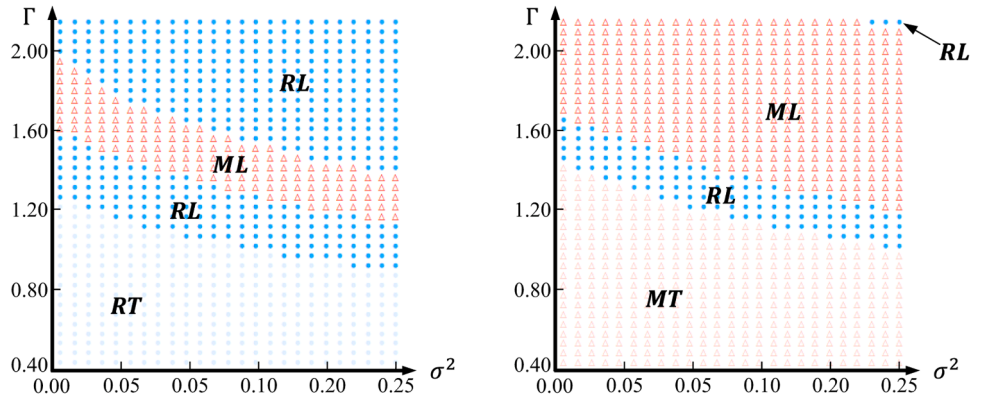
(a) $0 < \lambda \leq \frac{1}{4}$ (b) $\frac{1}{4} < \lambda < 1$

Fig. 3. Selling format strategy without supply risks.



(a) $0 < \lambda \leq \frac{1}{4}$ (b) $\frac{1}{4} < \lambda < 1$

Fig. 4. Selling format strategy with supply risks.



(a) $\lambda = 0.22$ ($< 1/4$) (b) $\lambda = 0.26$ ($> 1/4$)

Fig. 5. Selling format strategy under proportional live-streaming commission.

To better understand the results of Proposition 4, we depict the format and channel decisions with supply risks in Fig. 4. Comparing Fig. 3 and Fig. 4, we can see that even with supply risks, the e-tailer's format preference with low to moderate live-streaming efficiency (i.e., $\Gamma < \Gamma^M$ in Fig. 4 and $\Gamma < \frac{1}{1-\lambda}$ in Fig. 3) is clear and similar to the preference without supply risks. However, as the live-streaming efficiency increases, the closed-form thresholds for identifying the preference over the two selling formats are intractable. Thus, we further explore the e-tailer's preference by numerical examples.

In the numerical study, we set $\delta = 0.8$, $\alpha = 0.25$, and $\mu = 0.8$. In Fig. 5, the x-axis represents the variance σ^2, the y-axis represents the live-streaming efficiency Γ, and the two examples with $\lambda = 0.22$ and $\lambda = 0.26$ are selected to represent the two cases for $\lambda \leq 1/4$ and $\lambda > 1/4$, respectively. For each pair of (σ^2, Γ), we compare the e-tailer's profit under the reselling and marketplace formats, given the optimal channel decision as in Proposition 1, to derive which selling format generates a greater profit. We use different colors and shapes to differentiate the format. For each pair of parameters in Fig. 5, we use a blue dot to represent that the preferred format is reselling, and we use a red triangle to represent that the preferred outcome is the marketplace. We also use shades to differentiate whether or not to introduce a live-streaming channel, where the light blue/red indicates the cases without the live-streaming channel, and the dark blue/red shows the cases where a live-streaming channel is introduced. Fig. 5 shows a similar pattern to that in Fig. 3. That is, sufficiently high live-streaming efficiency may change the e-tailer's preference over the selling format, e.g., the marketplace is preferred for some cases in Fig. 5(a) and reselling is preferred for some cases in Fig. 5(b). Furthermore, we can observe that as the variance σ^2 increases, the thresholds on live-streaming efficiency for introducing a live-streaming channel decrease, and the thresholds to switch the selling format also decrease.

5. Extended models

5.1. The impact of live-streaming commission structures

In practice, the live-streaming commission may have different contractual forms, including proportional commission [2,27,40] and unit commission [54,58,59], and fixed commission [56,57]. The first two are streamers' sales effort based, and the last one is independent of streamers' performance. Given that the effort-based commission is more prominent in practice and this study analyzes the operational decisions by featuring streamers' sales effort, we focus on the first two commission structures. The proportional commission is studied in the main model, and we consider the unit commission in this extended model to investigate the robustness of commission structures. We use the symbol "∼" to

label all the parameters in this section.

Let r be the unit commission. In line with the proportional commission scenario, we consider the case where the unit commission is not excessively high to ensure positive demands and avoid trivial cases. Specifically, we assume that $r < \bar{r} = \min\left\{\frac{1}{\delta\theta^2}, \frac{1}{2-\delta+\delta^2\theta^2}\right\}$. This assumption is commonly adopted and consistent with the prior literature, such as Ji et al. [54], regarding the unit live-streaming commission. Then, under the reselling format, we have

$$\widetilde{\pi}_B^{RL}(w) = w(d_t + d_l), \tag{10}$$

$$\widetilde{\pi}_E^{RL}(p_t, p_l) = (p_t - w)d_t + (p_l - w - r)d_l. \tag{11}$$

Under the marketplace format, we consider $\lambda \le 1 - 2r$ so that the brand has a non-negative marginal. In this case, we have

$$\widetilde{\pi}_B^{ML}(p_t, p_l) = (1-\lambda)p_t d_t + ((1-\lambda)p_l - r)d_l. \tag{12}$$

In both cases, the streamer's profits are given by

$$\widetilde{\pi}_S^{ML}(e) = rd_l - \frac{1}{2}e^2. \tag{13}$$

We construct the analysis by following the same logic in Section 4. The optimal short-term decisions are summarized in the following two lemmas.

Lemma 4. *Under the reselling format, the streamer's optimal sales effort, the wholesale price, and the optimal retail prices are* $\widetilde{e}^{RL} = zr\delta\theta$, $\widetilde{w} = \frac{1-r\delta(1-z\delta\theta^2)}{2}$, *and*

$$(\widetilde{p}_t^{RL}, \widetilde{p}_l^{RL}) = \begin{cases} \left(\frac{3+r\delta(1-z\delta\theta^2)}{4}, \frac{3+r\delta(1-z\delta\theta^2+4z\theta^2)}{4}\right), & \Gamma < \frac{1}{z}, \\ \left(\frac{3-r\delta(1-z\delta\theta^2)}{4}, \frac{3+r(2-\delta+z\delta\theta^2(2+\delta))}{4}\right), & \Gamma \ge \frac{1}{z}. \end{cases}$$

Lemma 5. *Under the marketplace format,* $\widetilde{e}^{ML} = zr\delta\theta$, *and*

$$(\widetilde{p}_t^{ML}, \widetilde{p}_l^{ML}) = \begin{cases} \left(\frac{1-\lambda+r\delta(1-(1-\lambda)z\delta\theta^2)}{2(1-\lambda)}, \frac{1-\lambda+r(\delta+(2-\delta)(1-\lambda)z\delta\theta^2)}{2(1-\lambda)}\right), & \Gamma < \frac{1}{z(1-\lambda)}, \\ \left(\frac{1}{2}, \frac{(1-\lambda)(1+zr\delta\theta^2)+r}{2(1-\lambda)}\right), & \Gamma \ge \frac{1}{z(1-\lambda)}. \end{cases}$$

Lemmas 4 and 5 show the closed-form analytical expressions for the optimal short-term decisions with the unit live-streaming commission fee. Under each selling format, retail prices take different forms in different cases governed by live-streaming efficiency. Comparing the optimal short-term decisions with and without live streaming, we find the important neat results in the following corollary.

Corollary 3. *With the unit live-streaming commission, (i)* $\widetilde{p}_l^{RL} > \widetilde{p}_t^{RL} \ge \widetilde{p}_t^{RT}$, *and* $\widetilde{D}^{RL} < \widetilde{D}^{RT}$ *if and only if* $\Gamma < \frac{1}{z}$; *(ii)*$\widetilde{p}_l^{ML} > \widetilde{p}_t^{ML} \ge \widetilde{p}_t^{MT}$, *and* $\widetilde{D}^{ML} < \widetilde{D}^{MT}$ *if and only if* $\Gamma < \frac{1}{z(1-\lambda)}$; *(iii)* $\widetilde{e}^{RL} = \widetilde{e}^{ML}$.

Comparing Corollary 3 with Corollaries 1 and 2, we find that the live-streaming commission structure has diversified impacts on optimal short-term decisions. Recall that introducing a live-streaming channel with a proportional commission does not impact the retail price of the traditional channel and only causes an internal switch of the customers so that the total demand does not change. However, introducing a live-streaming channel with a unit commission does (weakly) increase the retail price of the traditional channel and may either increase or decrease the total demand depending on the live-streaming efficiency. Intuitively, the unit commission fee tends to increase the cost for the e-tailer/brand, which subsequently drives up product prices to offset the fees paid to the streamer. Since the introduction of a live-streaming channel (weakly) increases the retail price of the traditional channel, the total realized demand may decrease. However, if the live-streaming efficiency is sufficiently high, the streamer will increase the sales effort and finally lead to an increased total realized demand. Furthermore, the streamer's sales effort is affected by the selling format with the proportional commission but independent of the selling format with the unit commission. This is because the unit commission makes the streamer's profit independent of the revenue of the live-streaming channel.

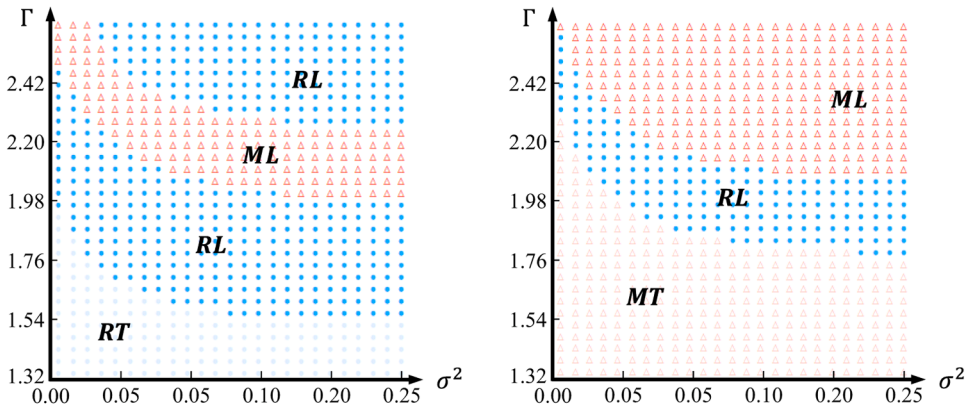
Based on the optimal short-term decisions, we can further analyze the channel and selling format choices. We first consider the special case with perfect supply viability.

Proposition 5. *Given perfect supply viability and a unit live-streaming commission fee, (i) it is more profitable to introduce a live-streaming channel if and only if* $\Gamma \ge 1$ *under the reselling format and* $\Gamma \ge \frac{1}{1-\lambda}$ *under the marketplace format; (ii) there exist two thresholds on the live-streaming efficiency* $\widetilde{\Gamma}_1$ *and* $\widetilde{\Gamma}_2$ *such that the e-tailer prefers the marketplace format if and only if either (a)* $0 < \lambda \le \frac{1}{4}$ *and* $\frac{1}{1-\lambda} \le \Gamma < \widetilde{\Gamma}_2$ *or (b)* $\frac{1}{4} < \lambda < 1$ *and* $\Gamma < \widetilde{\Gamma}_1$ *or* $\frac{1}{1-\lambda} \le \Gamma < \widetilde{\Gamma}_2$.

The comparison between Propositions 3 and 5 reveals that with perfect supply viability, the live-streaming commission structure does not influence the choice of live-streaming channel and the platform's selling format.

Next, we address the impact of random supply viability. Recall that the optimal short-term decisions vary across two scenarios based on the level of live-streaming efficiency; see Lemmas 4 and 5. The analytical complexity makes the theoretical analysis intractable. Therefore, we conduct extensive numerical studies to explore the impact of the supply viability on the channel and format decisions. In the numerical study, we set $\delta = 0.22$, and $r = 0.38$, and let the supply viability follow a normal distribution with mean $\mu = 1/2$. Similar to Fig. 5, we let the x-axis represent the variance σ^2, let the y-axis represent the live-streaming efficiency Γ, and select the two examples with $\lambda = 0.22$ and $\lambda = 0.26$ to represent the two cases for $\lambda \le 1/4$ and $\lambda > 1/4$. See Fig. 6 for the illustration. For each pair of (σ^2, Γ), we first calculate the expected profits under different scenarios and then conduct the comparison to reveal the e-tailer's preference. We use the same colors and shapes to mark the different preferences as used in Fig. 5.

We can observe that the outcomes in Fig. 6 are similar to the pattern in Fig. 5. For both cases in Fig. 6, there exist thresholds on the live-streaming efficiency such that introducing a live-streaming channel is more advantageous if the live-streaming efficiency is high, and the thresholds decrease in the variance. Regarding the e-tailer's format preference, as the live-streaming efficiency increases, Fig. 6(a) shows that the preferred format changes from reselling to the marketplace and then back to reselling, and Fig. 6(b) shows that the preferred format changes from marketplace to reselling and then back to the marketplace. Observations from Fig. 6 confirm the robustness of our main results in Section 4. Specifically, the introduction of a live-streaming channel and



(a) $\lambda = 0.22$ ($< 1/4$) (b) $\lambda = 0.26$ ($> 1/4$)

Fig. 6. Channel strategy and selling format strategy with the unit live-streaming commission.

the preference for the platform's selling format depend on live-streaming efficiency. This insight provides clear guidance for managers to enhance live-streaming efficiency.

Furthermore, we observe that under the unit live-streaming commission structure, the e-tailer's preference for the selling format is more sensitive when the variance in supply viability is low but less sensitive when the variance is high. While the streamer's sales effort is independent of the selling format under the unit commission, the platform and brand's profits vary significantly depending on the chosen selling format. If supply viability is less volatile, the e-commerce supply chain has a more predictable revenue outlook, allowing the platform to implement more refined selling formats to optimize income. In contrast, if the supply viability has a higher volatility, the players face the challenge of not being able to accurately forecast the match between supply and demand, and thus the platform is likely to prioritize managing supply chain risks over purely optimizing the selling format.

5.2. The impact of viability improvement

Supply viability improvement is a critical strategy for mitigating supply risks and enhancing the resilience and sustainability of supply chains [67]. Many firms invest significant resources in this area, using technologies such as artificial intelligence and digital supply chain [16, 22]. However, such improvements often come with considerable costs, creating a strategic challenge for firms weighing risk mitigation against the costs of improving supply viability. In this section, we extend our study to consider the viability improvement in the context of supply risks.

In the main model, we characterize supply risks by a random yield. Another way to feature supply risks is supply disruption, where the seller faces an all-or-nothing outcome. If a disruption occurs, production yields zero output, and the seller is unable to fulfill any orders. In this extended study, we restrict to this type of supply risks for tractability.

Specifically, we let β be the probability of a perfect yield, and $1-\beta$ be the probability of facing complete disruption. This is commonly adopted in studies about supply risks [21,23,68]. In line with Li et al. [23], we assume the seller can improve viability by increasing the success probability β, incurring a cost of $\frac{1}{2}\beta^2$. Here, we also refer to the success probability β as the supply viability level. In this section, we assume the supply viability level β is determined given the selection of selling format and channel decisions. For ease of presentation, we use the symbol "$\widehat{\ }$" to denote the functions and variables in this extended model.

Under the reselling format, we have

$$\widehat{\pi}_B^{RL}(w) = \beta w[(1-\delta)(1-p_t) + \delta(1-p_l+\theta e)], \tag{14}$$

$$\widehat{\pi}_E^{RL}(p_t,p_l) = \beta[(p_t-w)(1-\delta)(1-p_t) + (1-\alpha)(p_l-w)\delta(1-p_l+\theta e)] - \frac{1}{2}\beta^2, \tag{15}$$

$$\widehat{\pi}_S^{RL}(e) = \beta\left[\alpha(p_l-w)\delta(1-p_l+\theta e) - \frac{1}{2}e^2\right]. \tag{16}$$

Under the marketplace format, we have

$$\widehat{\pi}_B^{ML}(p_t,p_l) = \beta(1-\lambda)[p_t(1-\delta)(1-p_t) + (1-\alpha)p_l\delta(1-p_l+\theta e)] - \frac{1}{2}\beta^2, \tag{17}$$

$$\widehat{\pi}_S^{ML}(e) = \beta\left[\alpha(1-\lambda)p_l\delta(1-p_l+\theta e) - \frac{1}{2}e^2\right]. \tag{18}$$

Following the same logic as in Section 4, we first analyze the short-term decisions, including the streamer's sales effort, retail prices, and wholesale price, with the detailed results provided in the Appendix. We then optimize the expected profit over the supply viability level β, and the results are summarized in the following lemma.

Lemma 6. *(i) Under the reselling format, the e-tailer's optimal viability level is* $\widehat{\beta}^{RL} = \frac{1-\alpha\delta\left(1+(1-\delta)\theta^2\right)}{16\left(1-\alpha\delta\theta^2\right)}$*; and (ii) under the marketplace format, the brand's optimal viability level is* $\widehat{\beta}^{ML} = \frac{(1-\lambda)\left(1-\alpha\delta\left(1+(1-\lambda)(1-\delta)\theta^2\right)\right)}{4\left(1-(1-\lambda)\alpha\delta\theta^2\right)}$.

Lemmas 6 shows the optimal viability level under the two selling formats. We then examine its sensitivities regarding the key factors.

Corollary 4. *For $X = R, M$, we have (i) $\widehat{\beta}^{XL}$ decreases with α if and only if $\Gamma < \widehat{\Gamma}_1^X$; otherwise, it increases with α; (ii) $\widehat{\beta}^{XL}$ decreases in δ if and only if $\Gamma < \widehat{\Gamma}_2^X$; otherwise, it increases with δ. (iii) $\widehat{\beta}^{XL}$ increases with θ.*

Corollary 4 implies that if the live-streaming efficiency is low, allocating more revenue to the streamer decreases the sellers' profitability, hence reducing the motivation to improve supply viability. However, with high live-streaming efficiency, a greater commission to the streamer can incentivize increased sales efforts, making viability improvements more appealing for the sellers. Similarly, if the live-streaming efficiency is low, higher consumer awareness may not significantly drive sales, as the live-streaming channel remains ineffective, discouraging sellers from improving supply viability. Conversely, with high efficiency, increased consumer acceptance enhances sales potential, prompting sellers to improve supply viability. Lastly, the higher sales ability of the streamer directly enhances the live-streaming channel's performance, making viability improvements more valuable. As a result, we end up with: the better the streamer's ability, the higher

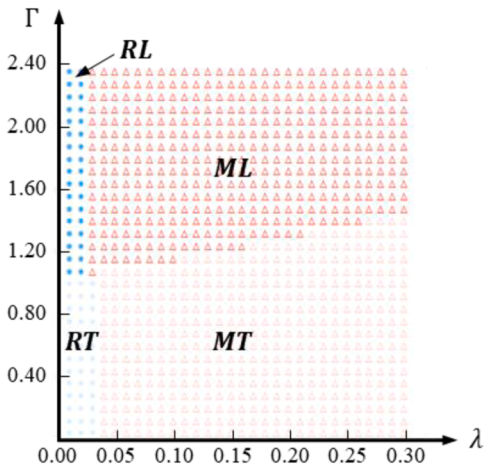


Fig. 7. Channel strategy and selling format strategy with viability improvement.

the returns on improving supply viability.

Next, by comparing the optimal sellers' profits under the reselling format ($\widehat{\pi}_E^{RT}$ and $\widehat{\pi}_E^{RL}$) and the marketplace format ($\widehat{\pi}_B^{MT}$ and $\widehat{\pi}_B^{ML}$). We show the decision on the introduction of a live-streaming channel in the following corollary.

Corollary 5. *$\widehat{\pi}_E^{RL} \geq \widehat{\pi}_E^{RT}$ if and only if $\Gamma \geq \widehat{\Gamma}_1^R$ under the reselling format, and $\widehat{\pi}_B^{ML} \geq \widehat{\pi}_B^{MT}$ if and only if $\Gamma \geq \widehat{\Gamma}_1^M$ under the marketplace format.*

Corollary 5 further confirms the robustness of our main findings in Section 4.2, demonstrating that live-streaming efficiency is the key factor for introducing a live-streaming channel.

Regarding the impact of supply viability improvement on the selling format choice, we conduct further numerical studies. We set $\delta = 0.8$, and $\alpha = 0.28$, which satisfy our modeling assumptions and illustrate the common case The x-axis represents the platform commission rate λ, while the y-axis represents live-streaming efficiency Γ, as shown in Fig. 7. We consider $\lambda \in (0.0.3)$, which is not excessively high as in real-world cases. For each pair of (λ, Γ), we highlight the different preferences as in Fig. 7. We can observe that as the platform commission rate increases, the preferred format shifts from the reselling to the marketplace.

6. Conclusion

In this study, we analytically investigate the role of live streaming in coping with e-commerce supply risks. We explore the channel decision on whether to introduce live streaming as well as the e-tailer's preference over the platform's selling format facing supply risks. The following important managerial insights are identified.

Supply chain viability: Live streaming plays a positive role in maintaining supply chain viability by enhancing the supply chain's overall ability to address supply risks, given that the expected supply viability level is not too low. Although supply risks typically harm supply chain performance, it is interesting to find that the adjustment power of the live streaming channel can help mitigate the negative effect.

Incorporating live streaming: It is wise to incorporate live streaming into the operations when live-streaming efficiency is sufficiently high. The corresponding condition (i.e., threshold) in fact is jointly determined by the customers' acceptance level of shopping from the live-streaming channel and the streamer's ability to sell. The threshold decreases in the variance of the supply viability, implying that incorporating live streaming is crucially important for the e-commerce supply chains facing supply risks with high volatility.

The impact of platform's selling format: The reselling format requires a lower threshold on live-streaming efficiency so that it is optimal to incorporate the live-streaming channel. This finding shows that the reselling format exhibits a higher acceptance of live streaming than the marketplace format. The better support for the new normal via adopting the reselling format will change the e-tailer's traditional preference over the two selling formats, which solely depends on the platform's commission rate. Furthermore, the combined effect of live streaming and supply risks further influences the preference for the selling format.

This paper has a non-trivial limitation with respect to the assumption that the proportion of customers aware of live streaming is exogenously given. This is, in fact, quite common in the literature and supported by some practices. However, it is also true that e-commerce platforms may exert marketing efforts to attract more consumers to visit or invest in digital technologies to improve the market power of live streaming [69, 70]. All these measures may mean that the proportion of customers aware of live streaming can be affected by e-commerce platforms. We postpone this extended study to future research. Future research could also incorporate demand risks, providing a more comprehensive analysis of how live streaming can mitigate risks on both the supply and demand sides of the supply chain. Additionally, exploring more sophisticated consumer behavior like socially responsible conscious [71], fairness concerns [72,73] under live-streaming e-commerce can also be interesting future research potential directions.

CRediT authorship contribution statement

Qingying Li: Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis. **Shuo Zhu:** Writing – original draft, Investigation, Writing – review & editing. **Tsan-Ming Choi:** Conceptualization, Writing – review & editing, Supervision. **Bin Shen:** Writing – original draft, Supervision, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Proof of Lemma 1. The optimal solutions can be obtained by backwards induction, following the event sequence depicted in Fig. 2. We first analyze the optimal sales effort given the wholesale price w and the retail prices p_t, p_l. From (3), we have:

$$\pi_S^{RL}(e)=\begin{cases}0,\ if\ e<\dfrac{p_l-p_t}{\theta},\\ \alpha(p_l-w)z\delta(1-p_l+\theta e)-\dfrac{1}{2}e^2,\ if\ \dfrac{p_l-p_t}{\theta}\le e<\dfrac{p_l}{\theta},\\ \alpha(p_l-w)z\delta-\dfrac{1}{2}e^2,\ if\ e\ge\dfrac{p_l}{\theta}.\end{cases}$$

It suffices to consider the last two cases. If $\frac{p_l-p_t}{\theta}\le e<\frac{p_l}{\theta}$, we can check that $\frac{d^2\pi_S^{RL}(e)}{de^2}=-1<0$, which implies the concavity of $\pi_S^{RL}(e)$. By solving the first-order condition (FOC), we have $e=z\alpha\delta\theta(p_l-w)$, which can be either smaller than $\frac{p_l-p_t}{\theta}$ or larger than $\frac{p_l}{\theta}$. Given that $\alpha<\frac{2}{3\delta\theta^2}$, we compare the two critical values. Thus, we have the local maxima over $\frac{p_l-p_t}{\theta}\le e<\frac{p_l}{\theta}$ as follows:

$$e^{RL}(p_t,p_l,w)=\begin{cases}z\alpha\delta\theta(p_l-w),\ if\ p_l<\dfrac{p_t-z\alpha\delta\theta^2w}{1-z\alpha\delta\theta^2},\\ \dfrac{p_l-p_t}{\theta},\ if\ p_l\ge\dfrac{p_t-z\alpha\delta\theta^2w}{1-z\alpha\delta\theta^2}.\end{cases}\tag{A.1}$$

If $e\ge\frac{p_l}{\theta}$, it is easy to check that $\pi_S^{RL}(e)$ decreases in e. Thus, the local maxima over $e\ge\frac{p_l}{\theta}$ is $\frac{p_l}{\theta}$. Comparing the streamer's profits at the two local maxima, we can check the streamer's optimal sales effort is attained at (A.1).

Next, we derive the e-tailer's optimal retail prices. Regarding the second case in (A.1), by putting $e^{RL}(p_t,p_l,w)$ to (2), we have:

$$\pi_E^{RL}(p_t,p_l)=(p_t-w)z(1-\delta)(1-p_t)+(1-\alpha)(p_l-w)z\delta(1-p_t),$$

where $\frac{\partial\pi_E^{RL}(p_t,p_l)}{\partial p_l}>0$. This implies the e-tailer will choose a sufficiently high p_l, which leads to "zero profit" for the streamer and hence cannot be optimal. Consider the first case in (A.1). If $p_l<\frac{p_t-z\alpha\delta\theta^2w}{1-z\alpha\delta\theta^2}$, putting $e^{RL}(p_t,p_l,w)$ to (2), we have:

$$\pi_E^{RL}(p_t,p_l)=(p_t-w)z(1-\delta)(1-p_t)+(1-\alpha)(p_l-w)z\delta\big(1-\big(1-z\alpha\delta\theta^2\big)p_l-z\alpha\delta\theta^2w\big).\tag{A.2}$$

By checking that $\frac{\partial^2\pi_E^{RL}(p_t,p_l)}{\partial p_l^2}=-2z(1-\alpha)\delta\big(1-z\alpha\delta\theta^2\big)<0$, we obtain the e-tailer's optimal live-streaming price given the traditional retail price as $p_l^{RL}(p_t)=\min\left\{\frac{1+w-2z\alpha\delta\theta^2w}{2\left(1-z\alpha\delta\theta^2\right)},\ \frac{p_t-z\alpha\delta\theta^2w}{1-z\alpha\delta\theta^2}\right\}$, or specifically:

$$p_l^{RL}(p_t)=\begin{cases}\dfrac{p_t-z\alpha\delta\theta^2w}{1-z\alpha\delta\theta^2},\ if\ p_t\le\dfrac{1+w}{2},\\ \dfrac{1+w-2z\alpha\delta\theta^2w}{2\left(1-z\alpha\delta\theta^2\right)},\ if\ p_t>\dfrac{1+w}{2}.\end{cases}\tag{A.3}$$

Putting (A.3) to (A.2), we have:

$$\pi_E^{RL}(p_t)=\begin{cases}(p_t-w)z(1-\delta)(1-p_t)+\dfrac{z\delta(1-\alpha)(p_t-w)(1-p_t)}{1-z\alpha\delta\theta^2},\ if\ p_t\le\dfrac{1+w}{2},\\ (p_t-w)z(1-\delta)(1-p_t)+\dfrac{z\delta(1-\alpha)(1-w)^2}{4\left(1-z\alpha\delta\theta^2\right)},\ if\ p_t>\dfrac{1+w}{2}.\end{cases}\tag{A.4}$$

It is easy to check from (A.4) that $\pi_E^{RL}(p_t)$ is concave in each case. From FOC, we obtain $p_t^{RL}(w)=\frac{1+w}{2}$. Therefore, we have $p_l^{RL}(w)=\frac{1+w-2z\alpha\delta\theta^2w}{2\left(1-z\alpha\delta\theta^2\right)}$, $e^{RL}(w)=\frac{(1-w)z\alpha\delta\theta}{2\left(1-z\alpha\delta\theta^2\right)}$.

Consider the brand's optimal wholesale price. By putting the retail prices and the sales effort into (1), we have $\pi_B^{RL}(w)=\frac{zw(1-w)}{2}$. It is easy to check that the optimal wholesale price is $w^{RL}=\frac{1}{2}$. Putting $w^{RL}=\frac{1}{2}$ into $p_t^{RL}(w)$, $p_l^{RL}(w)$ and $e^{RL}(w)$ derives Lemma 1. ▌

Proof of Lemma 2. Similar to the proof of Lemma 1, we first analyze the optimal sales effort given the retail prices p_t, p_l. From (5), we have:

$$\pi_S^{ML}(e)=\begin{cases}0,\ if\ e<\dfrac{p_l-p_t}{\theta},\\ \alpha(1-\lambda)p_lz\delta(1-p_l+\theta e)-\dfrac{1}{2}e^2,\ if\ \dfrac{p_l-p_t}{\theta}\le e<\dfrac{p_l}{\theta},\\ \alpha(1-\lambda)p_lz\delta-\dfrac{1}{2}e^2,\ if\ e\ge\dfrac{p_l}{\theta}.\end{cases}$$

It suffices to consider the last two cases. If $\frac{p_l-p_t}{\theta}\le e<\frac{p_l}{\theta}$, we can check that $\frac{d^2\pi_S^{ML}(e)}{de^2}=-1<0$, which implies the concavity of $\pi_S^{ML}(e)$. By solving the FOC, we have $e=z\alpha\delta\theta(1-\lambda)p_l$, which can be either smaller than $\frac{p_l-p_t}{\theta}$ or larger than $\frac{p_l}{\theta}$. Given that $\alpha<\frac{2}{3\delta\theta^2}$, we compare the two critical values and derive the local maxima over $\frac{p_l-p_t}{\theta}\le e<\frac{p_l}{\theta}$ as follows:

$$e^{ML}(p_t,p_l)=\begin{cases} z\alpha\delta\theta(1-\lambda)p_l, \ if\ p_l<\dfrac{p_t}{1-(1-\lambda)z\alpha\delta\theta^2}, \\ \dfrac{p_l-p_t}{\theta}, \ if\ p_l\geq\dfrac{p_t}{1-(1-\lambda)z\alpha\delta\theta^2}. \end{cases} \tag{A.5}$$

If $e\geq\frac{p_l}{\theta}$, it is easy to check that $\pi_S^{ML}(e)$ decreases in e. Thus, the local maxima over $e\geq\frac{p_l}{\theta}$ is $\frac{p_l}{\theta}$. Comparing the streamer's profits at the two local maxima, we can show that the streamer's optimal sales effort is attained at (A.5).

Next, we derive the brand's optimal retail prices. Regarding the second case in (A.5), by putting $e^{ML}(p_t,p_l)$ to (4), we have:

$$\pi_B^{ML}(p_t,p_l)=(1-\lambda)(p_tz(1-\delta)(1-p_t)+(1-\alpha)p_lz\delta(1-p_t)),$$

where $\frac{\partial\pi_B^{ML}(p_t,p_l)}{\partial p_l}>0$. This implies the brand will choose a sufficiently high p_l, which leads to "zero profit" for the streamer and hence cannot be optimal. Consider the first case in (A.5). If $p_l<\frac{p_t}{1-(1-\lambda)z\alpha\delta\theta^2}$, we have:

$$\pi_B^{ML}(p_t,p_l)=(1-\lambda)p_tz(1-\delta)(1-p_t)+(1-\lambda)(1-\alpha)p_lz\delta\big(1-\big(1-(1-\lambda)z\alpha\delta\theta^2\big)p_l\big). \tag{A.6}$$

Noting that $\frac{\partial^2\pi_B^{ML}(p_t,p_l)}{\partial p_l^2}=-2\delta z(1-\alpha)(1-\lambda)\big(1-(1-\lambda)z\alpha\delta\theta^2\big)<0$, we can easily obtain the brand's optimal live-streaming price given the traditional retail price as $p_l^{ML}(p_t)=\min\left\{\frac{1}{2\left(1-(1-\lambda)z\alpha\delta\theta^2\right)},\frac{p_t}{1-(1-\lambda)z\alpha\delta\theta^2}\right\}$, or specifically:

$$p_l^{ML}(p_t)=\begin{cases} \dfrac{p_t}{1-(1-\lambda)z\alpha\delta\theta^2}, \ if\ p_t\leq\dfrac{1}{2}, \\ \dfrac{1}{2\big(1-(1-\lambda)z\alpha\delta\theta^2\big)}, \ if\ p_t>\dfrac{1}{2}. \end{cases} \tag{A.7}$$

Substituting (A.7) to (A.6), we have:

$$\pi_B^{ML}(p_t)=\begin{cases} (1-\lambda)\left(p_tz(1-\delta)(1-p_t)+\dfrac{(1-\alpha)z\delta p_t(1-p_t)}{1-(1-\lambda)z\alpha\delta\theta^2}\right), \ if\ p_t\leq\dfrac{1}{2}, \\ (1-\lambda)\left(p_tz(1-\delta)(1-p_t)+\dfrac{(1-\alpha)z\delta}{4\big(1-(1-\lambda)z\alpha\delta\theta^2\big)}\right), \ if\ p_t>\dfrac{1}{2}. \end{cases}$$

It is easy to check that $\pi_B^{ML}(p_t)$ is concave in each case. From FOC, we obtain $p_t^{ML}=\frac{1}{2}$, putting which into $p_l^{ML}(p_t)$ and $e^{ML}(p_t,p_l)$ derives Lemma 2. ▌

Proof of Corollary 1. From $\delta\theta^2<\frac{2}{3\alpha}$, we can check $1-z\alpha\delta\theta^2>0$ and $1-(1-\lambda)z\alpha\delta\theta^2>0$. Thus, from Lemma 1 and Lemma 2, we have $p_l^{RL}-p_t^{RL}=\frac{z\alpha\delta\theta^2}{4\left(1-z\alpha\delta\theta^2\right)}>0$ and $p_l^{ML}-p_t^{ML}=\frac{(1-\lambda)z\alpha\delta\theta^2}{2\left(1-(1-\lambda)z\alpha\delta\theta^2\right)}>0$.

Putting the equilibrium solutions in Lemma 1 and Lemma 2 to the total demand functions, we have $D^{RL}=D^{RT}=\frac{z}{4}$ and $D^{ML}=D^{MT}=\frac{z}{2}$. ▌

Proof of Corollary 2. From Lemma 1 and Lemma 2, we have $p_t^{ML}-p_t^{RL}=-1/4$, and $D^{ML}-D^{RL}=z/4$. The results imply that $p_t^{ML}<p_t^{RL}$ and $D^{ML}>D^{RL}$. Regarding the live-streaming prices, we have:

$$p_l^{ML}-p_l^{RL}=\frac{z\alpha(1-\lambda)\left(-2z\alpha\left(\delta\theta^2\right)^2+3\delta\theta^2\right)-1}{4\big(1-z\alpha(1-\lambda)\delta\theta^2\big)\big(1-z\alpha\delta\theta^2\big)},$$

where the denominator is positive and the sign depends on the numerator. Let $f(x)=z\alpha(1-\lambda)\left(-2z\alpha x^2+3x\right)-1$. We can check $f(x)$ is concave. If $\lambda\geq1/9$, $p_l^{ML}-p_l^{RL}<0$. If $\lambda<1/9$, there exist two zero roots to $f(x)=0$, and the greater zero root is greater than $\frac{2}{3\alpha}$. In this case, we let Γ_0 be either the smaller zero root to $f(x)=0$ or the upper bound $\frac{2}{3\alpha}$, whichever is smaller. Then, we have $p_l^{ML}>p_l^{RL}$ if and only if $\lambda<1/9$ and $\Gamma>\Gamma_0$.

Regarding the optimal sales effort, we have:

$$e^{ML}-e^{RL}=\frac{z\alpha\delta\theta\left[-(1-\lambda)z\alpha\delta\theta^2+1-2\lambda\right]}{4\big(1-z\alpha\delta\theta^2\big)\big(1-z\alpha\delta\theta^2(1-\lambda)\big)},$$

where the denominator is positive and the sign depends on the numerator. It is easy to check that the term in the bracket of the numerator decreases in Γ (i.e., $\delta\theta^2$) and the zero root is attained at $\frac{1-2\lambda}{z\alpha(1-\lambda)}$. If $0<\lambda<\frac{1}{9}$, we have $\frac{1-2\lambda}{z\alpha(1-\lambda)}>\frac{2}{3\alpha}$. Thus, we have $e^{ML}-e^{RL}>0$.

To summarize, Corollary 2 is proven. ▌

Proof of Lemma 3 We first list the optimal profits without supply risks, and the equilibrium solutions without introducing a live-streaming channel.

Table A.1
The optimal profits without supply risks.

	$\overline{\Pi}_E^{XT}$	$\overline{\Pi}_B^{XT}$	$\overline{\Pi}_E^{XL}$	$\overline{\Pi}_B^{XL}$
Reselling format	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1-\alpha\delta(1+(1-\delta)\theta^2)}{16(1-\alpha\delta\theta^2)}$	$\frac{1}{8}$
Marketplace format	$\frac{\lambda}{4}$	$\frac{1-\lambda}{4}$	$\frac{\lambda(1-(1-\lambda)(1-\delta)\alpha\delta\theta^2)}{4(1-(1-\lambda)\alpha\delta\theta^2)}$	$\frac{(1-\lambda)(1-\alpha\delta(1+(1-\lambda)(1-\delta)\theta^2))}{4(1-(1-\lambda)\alpha\delta\theta^2)}$

Table A.2
The equilibrium solutions with a traditional channel.

	w^{RT}	p_t^{XT}	D^{XT}	Π_E^{XT}	Π_B^{XT}
Reselling format	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{z}{4}$	$\frac{\mu}{16}$	$\frac{\mu}{8}$
Marketplace format	–	$\frac{1}{2}$	$\frac{z}{2}$	$\frac{\mu\lambda}{4}$	$\frac{\mu(1-\lambda)}{4}$

(i) From Table A.1 and Table A.2, we have: $\Pi_E^{RT}-\overline{\Pi}_E^{RT}=\frac{\mu-1}{16}<0$, $\Pi_B^{RT}-\overline{\Pi}_B^{RT}=\frac{\mu-1}{8}<0$, $\Pi_E^{MT}-\overline{\Pi}_E^{MT}=\frac{\lambda(\mu-1)}{4}<0$, and $\Pi_B^{MT}-\overline{\Pi}_B^{MT}=\frac{(1-\lambda)(\mu-1)}{4}<0$.
(ii) From Table A.1 and (6), we have:

$$\Pi_E^{RL}-\overline{\Pi}_E^{RL}=\frac{(1-\alpha)\alpha\delta^2\theta^2\left[\sigma^2-\frac{(1-\mu)\left(1-\mu\alpha\delta\theta^2\right)^2\left((1-\delta)\left(1-\alpha\delta\theta^2\right)\left(1-\alpha\delta\theta^2\mu\right)+\delta(1-\alpha)\right)}{(1-\alpha)\alpha\delta^2\theta^2\left(1-\alpha\delta\theta^2\right)}\right]}{16\left(1-\alpha\delta\theta^2\mu\right)^3},$$

the sign of which depends on the terms in the bracket of the numerator. Denote $g(x)=\sigma^2-\frac{(1-x)\left(1-\alpha\delta\theta^2x\right)^2\left((1-\delta)\left(1-\alpha\delta\theta^2\right)\left(1-\alpha\delta\theta^2x\right)+(1-\alpha)\delta\right)}{(1-\alpha)\alpha\delta^2\theta^2\left(1-\alpha\delta\theta^2\right)}$. It is easy to check that $g(x)$ increases in x, $g(0)<0$ and $g(1)>0$. Let μ^R be the unique solution to $g(x)=0$. We have $\Pi_E^{RL}<\overline{\Pi}_E^{RL}$ if and only if $\mu<\mu^R$. Since $g(x)$ increases in σ^2, the zero root μ^R decreases in σ^2.

From Table A.1 and (9), we also have:

$$\Pi_B^{ML}-\overline{\Pi}_B^{ML}=\frac{(1-\alpha)(1-\lambda)^2\alpha\delta^2\theta^2\left[\sigma^2-\frac{(1-\mu)\left(1-(1-\lambda)\mu\alpha\delta\theta^2\right)^2\left((1-\delta)\left(1-(1-\lambda)\alpha\delta\theta^2\right)\left(1-(1-\lambda)\mu\alpha\delta\theta^2\right)+\delta(1-\alpha)\right)}{\left(1-(1-\lambda)\alpha\delta\theta^2\right)(1-\alpha)(1-\lambda)\alpha\delta^2\theta^2}\right]}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)^3},$$

the sign of which depends on the terms in the bracket of the numerator. Denote $h(x)=\sigma^2-\frac{(1-x)\left(1-\alpha\delta\theta^2(1-\lambda)x\right)^2\left((1-\delta)\left(1-(1-\lambda)\alpha\delta\theta^2\right)\left(1-(1-\lambda)\alpha\delta\theta^2x\right)+(1-\alpha)\delta\right)}{(1-\alpha)(1-\lambda)\alpha\delta^2\theta^2\left(1-(1-\lambda)\alpha\delta\theta^2\right)}$. It is easy to check that $h(x)$ increases in x, $h(0)<0$ and $h(1)>0$. Let μ^M be the unique solution to $h(x)=0$. We have $\Pi_B^{ML}-\overline{\Pi}_B^{ML}<0$ if and only if $\mu<\mu^M$. Similarly, since $h(x)$ increases in σ^2, the zero root μ^R decreases in σ^2. ▌

Proofs of Propositions 1 and 2 From (6) and Table A.2, we have:

$$\Pi_E^{RL}-\Pi_E^{RT}=\frac{(1-\alpha)\alpha\delta^2\theta^2\left[\sigma^2-\frac{\mu\left(1-\mu\delta\theta^2\right)\left(1-\alpha\mu\delta\theta^2\right)^2}{(1-\alpha)\delta\theta^2}\right]}{16\left(1-\alpha\mu\delta\theta^2\right)^3},$$

the sign of which depends on the term in the bracket of the numerator. Denote $G^R(x)=\sigma^2-\frac{\mu(1-\mu x)(1-\alpha\mu x)^2}{(1-\alpha)x}$. It is easy to check that if $x\geq\frac{1}{\mu}$, we have $G^R(x)>0$ and thus $\Pi_E^{RL}-\Pi_E^{RT}>0$. Otherwise, we can check that $G^R(x)$ increases in x, $G^R(0^+)<0$, and $G^R(1/\mu)>0$. Thus, there exists a unique solution, denoted as Γ^R, to $G^R(x)=0$ over $x\in(0,1/\mu)$. Then, we have $\Pi_E^{RL}\geq\Pi_E^{RT}$ if and only if $\Gamma\geq\Gamma^R$.

From (9) and Table A.2, we also have:

$$\Pi_B^{ML}-\Pi_B^{MT}=\frac{(1-\alpha)\alpha\delta^2\theta^2(1-\lambda)^2\left[\sigma^2-\frac{\mu\left(1-(1-\lambda)\mu\delta\theta^2\right)\left(1-(1-\lambda)\alpha\mu\delta\theta^2\right)^2}{(1-\alpha)(1-\lambda)\delta\theta^2}\right]}{4\left(1-\alpha\delta\theta^2(1-\lambda)\mu\right)^3},$$

the sign of which depends on the term in the bracket of the numerator. Denote $G^M(x)=\sigma^2-\frac{\mu(1-(1-\lambda)\mu x)(1-(1-\lambda)\alpha\mu x)^2}{(1-\alpha)(1-\lambda)x}$. If $x\geq\frac{1}{\mu(1-\lambda)}$, it is easy to show that $G^M(x)>0$ and $\Pi_B^{ML}-\Pi_B^{MT}>0$. Otherwise, we can check that $G^M(x)$ increases in x, $G^M(0^+)<0$, and $G^M\left(\frac{1}{\mu(1-\lambda)}\right)>0$. Thus, there exists a unique solution, denoted as Γ^M, to $G^M(x)=0$ over $x\in\left(0,\frac{1}{\mu(1-\lambda)}\right)$. Then, we have $\Pi_B^{ML}\geq\Pi_B^{MT}$ if and only if $\Gamma\geq\Gamma^M$.

Recall that $G^R(x)$ and $G^M(x)$ are both increasing functions. We have $\Gamma^R < \Gamma^M$ from $G^R(x) > G^M(x)$ and Γ^X decreases in σ^2 from $G^X(x)$ increases in σ^2, where $X = R, M$. Regarding Γ^M's sensitivity with λ, it is easy to check that $\frac{\partial \Gamma^M}{\partial \lambda} = \frac{\Gamma^M}{1-\lambda} > 0$. ▌

Proof of Proposition 3. Without supply risks, the thresholds on introducing a live-streaming channel in Proposition 1 can be simplified as $\Gamma^R = 1$ and $\Gamma^M = \frac{1}{1-\lambda}$. Thus, we establish the proof by considering the value of Γ in three cases.

Case (i) $\Gamma < 1$. In this case, neither the e-tailer nor the brand introduces a live-streaming channel, and $\overline{\Pi}_E^{RT} - \overline{\Pi}_E^{MT} = \frac{1-4\lambda}{16}$. Thus, we have $\overline{\Pi}_E^{RT} - \overline{\Pi}_E^{MT} \geq 0$ if $\lambda \leq \frac{1}{4}$, and $\overline{\Pi}_E^{RT} - \overline{\Pi}_E^{MT} < 0$ if $\lambda > \frac{1}{4}$.

Case (ii) $1 \leq \Gamma < \frac{1}{1-\lambda}$. In this case, the e-tailer introduces a live-streaming channel but the brand does not, and $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{MT} = \frac{(4\lambda-1+\delta)\alpha\delta\theta^2+1-4\lambda-\alpha\delta}{16\left(1-\alpha\delta\theta^2\right)}$, the sign of which depends on the numerator. Let $j(x) = (4\lambda - 1 + \delta)\alpha x + 1 - 4\lambda - \alpha\delta$, which satisfies $j(1) = (1 - 4\lambda)(1 - \alpha)$, and $j\left(\frac{1}{1-\lambda}\right) = \frac{(1-4\lambda)(1-\lambda-\alpha)+\lambda\alpha\delta}{1-\lambda}$. We then evaluate the sign of $j(x)$ according to the value of λ. If $\lambda \leq \frac{1}{4}$, we can check that $j(1) \geq 0$ and $j\left(\frac{1}{1-\lambda}\right) > 0$. This, together with the monotonicity of $j(x)$ imply that $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{MT} \geq 0$. If $\lambda > \frac{1}{4}$, we can check that $j(x)$ increases, $j(1) < 0$, and $j\left(\frac{1}{1-\lambda}\right)$ can be either greater or smaller than 0. Let Γ_1' be the unique solution to $j(x) = 0$ and let $\Gamma_1 = \min\left\{\Gamma_1', \frac{1}{1-\lambda}\right\}$. Then, we have $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{MT} < 0$ if $\lambda > \frac{1}{4}$ and $1 \leq \Gamma < \Gamma_1$.

Case (iii) $\Gamma \geq \frac{1}{1-\lambda}$. In this case, both the e-tailer and brand introduce a live-streaming channel, and we have:

$$\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} = \frac{A\left(\delta\theta^2\right)^2 + B\delta\theta^2 + C}{16\left(1-\alpha\delta\theta^2\right)\left(1-(1-\lambda)\alpha\delta\theta^2\right)},$$

where $A = \alpha^2(1-4\lambda)(1-\delta)(1-\lambda)$, $B = -\alpha\left(4(1-\delta)\lambda^2 - (9-4\delta-\alpha\delta)\lambda + 2 - \alpha\delta - \delta\right)$, $C = 1 - \alpha\delta - 4\lambda$, and the denominator is positive. Let $k(x) = Ax^2 + Bx + C$. We discuss the sign of $k(x)$ according to the value of λ. (a) If $\lambda < \frac{1}{4}$, then $A > 0$ and $k(x)$ is convex. If $k(x)$ has no root or only one root, then $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} > 0$, and in this case, we let $\Gamma_2 = \frac{1}{1-\lambda}$. Otherwise, we let $\Gamma'$and Γ'', where $\Gamma' < \Gamma''$, be the two roots to $k(x) = 0$. We can check that either $\Gamma' < \frac{1}{1-\lambda}$ or $\Gamma' > \frac{2}{3\alpha}$. We then specify the value of Γ_2 for different cases of zero roots. If $\Gamma'' \leq \frac{1}{1-\lambda}$ or $\Gamma' > \frac{2}{3\alpha}$, then $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} \geq 0$ over $\Gamma \in \left[\frac{1}{1-\lambda}, \frac{2}{3\alpha}\right]$, and we let $\Gamma_2 = \frac{1}{1-\lambda}$. If $\Gamma' < \frac{1}{1-\lambda} < \Gamma'' < \frac{2}{3\alpha}$, then $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} < 0$ if and only if $\Gamma < \Gamma''$, and we let $\Gamma_2 = \Gamma''$. If $\Gamma' < \frac{1}{1-\lambda}$ and $\Gamma'' \geq \frac{2}{3\alpha}$, then $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} < 0$ over $\Gamma \in \left[\frac{1}{1-\lambda}, \frac{2}{3\alpha}\right)$, and we let $\Gamma_2 = \frac{2}{3\alpha}$. Summarizing the subcases above, we have: $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} < 0$ if and only if $\Gamma < \Gamma_2$. (b) If $\lambda = \frac{1}{4}$, then $A = 0$, $B > 0$, and $k(x)$ increases. It is easy to check that $k\left(\frac{1}{1-\lambda}\right) < 0$. Let $\Gamma_2{}'$ be the unique solution to $k(x) = 0$ and $\Gamma_2 = \min\left\{\Gamma_2{}', \frac{2}{3\alpha}\right\}$. Then, we have $\overline{\Pi}_E^{RL} - \overline{\Pi}_E^{ML} < 0$ if and only if $\Gamma < \Gamma_2$. (c) If $\lambda > \frac{1}{4}$, then we can show that $A < 0$, $k(x)$ is concave and there exist two zero roots. In this case, we let Γ' and Γ'', $\frac{1}{1-\lambda} < \Gamma' < \Gamma''$, be the two roots to $k(x) = 0$. It is easy to check that $k\left(\frac{1}{1-\lambda}\right) < 0$, and $\Gamma'' > \frac{2}{3\alpha}$. In this case, we let $\Gamma_2 = \min\left\{\Gamma', \frac{2}{3\alpha}\right\}$. Then, $\overline{\Pi}_E^{RL} < \overline{\Pi}_E^{ML}$ if and only if $\Gamma < \Gamma_2$. Summarizing the three subcases in case (iii), we identify the threshold Γ_2 such that $\overline{\Pi}_E^{RL} < \overline{\Pi}_E^{ML}$ if and only if $\frac{1}{1-\lambda} \leq \Gamma < \Gamma_2$.

To summarize, Proposition 3 is proven. ▌

Proof of Proposition 4. If $\Gamma < \Gamma^R$, then neither the e-tailer nor the brand introduces a live-streaming channel, and $\Pi_E^{RT} - \Pi_E^{MT} = \frac{\mu(1-4\lambda)}{16}$. It is easy to verify that $\Pi_E^{RT} \geq \Pi_E^{MT}$ if $\lambda \leq 1/4$, and $\Pi_E^{RT} < \Pi_E^{MT}$ if $\lambda > 1/4$. If $\Gamma^R \leq \Gamma < \Gamma^M$, then the e-tailer introduces a live-streaming channel but the brand does not, and we have:

$$\Pi_E^{RL} - \Pi_E^{MT} = \frac{\mu((1-\delta)-4\lambda)\left(1-\alpha\delta\theta^2\mu\right)^3 + (1-\alpha)\delta\mu\left(1-\alpha\delta\theta^2\mu\right)^2 + (1-\alpha)\alpha\delta^2\theta^2\sigma^2}{16\left(1-\alpha\delta\theta^2\mu\right)^3},$$

which is positive if $\lambda \leq 1/4$. ▌

Proof of Lemma 4. The optimal solutions can be obtained by backwards induction according to the event sequence depicted in Fig. 2. We first analyze the streamer's optimal sales effort. From (13), we have:

$$\tilde{\pi}_S^{XL}(e) = \begin{cases} 0, \ if \ e < \dfrac{p_l - p_t}{\theta}, \\ zr\delta(1 - p_l + \theta e) - \dfrac{1}{2}e^2, \ if \ \dfrac{p_l - p_t}{\theta} \leq e < \dfrac{p_l}{\theta}, \\ zr\delta - \dfrac{1}{2}e^2, \ if \ e \geq \dfrac{p_l}{\theta}. \end{cases}$$

It suffices to consider the last two cases. If $\frac{p_l - p_t}{\theta} \leq e < \frac{p_l}{\theta}$, we can check that $\frac{d^2\tilde{\pi}_S^{XL}(e)}{de^2} = -1 < 0$, which implies the concavity of $\tilde{\pi}_S^{XL}(e)$. By solving the first-order condition (FOC), we have $e = zr\delta\theta$, which can be either smaller than $\frac{p_l - p_t}{\theta}$ or larger than $\frac{p_l}{\theta}$. Comparing the two critical values, we have the local maxima over $\frac{p_l - p_t}{\theta} \leq e < \frac{p_l}{\theta}$ as follows:

$$\widetilde{e}^{XL}(p_t,p_l)=\begin{cases}\dfrac{p_l}{\theta},\ if\ p_l<zr\delta\theta^2,\\ zr\delta\theta,\ if\ zr\delta\theta^2\le p_l<p_t+zr\delta\theta^2,\\ \dfrac{p_l-p_t}{\theta},\ if\ p_l\ge p_t+zr\delta\theta^2.\end{cases}\tag{A.8}$$

If $e\ge\frac{p_l}{\theta}$, it is easy to check that $\widetilde{\pi}_S^{XL}(e)$ decreases in e. Thus, the local maxima over $e\ge\frac{p_l}{\theta}$ is $\frac{p_l}{\theta}$. Comparing the streamer's profits at the two local maxima, we can check the streamer's optimal sales effort is attained at (A.8).

Next, we derive the e-tailer's optimal retail prices. Regarding the third case in (A.8), by putting $\widetilde{e}^{RL}(p_t,p_l)$ to (11) we have:

$$\widetilde{\pi}_E^{RL}(p_t,p_l)=(p_t-w)z(1-\delta)(1-p_t)+(p_l-w-r)z\delta(1-p_t),$$

where $\frac{\partial\widetilde{\pi}_E^{RL}(p_t,p_l)}{\partial p_l}>0$. This implies the e-tailer will choose a sufficiently high p_l, which leads to "zero profit" for the streamer and hence cannot be optimal. Thus, it suffices to consider the first two cases in (A.8). Consider the second case in (A.8). If $zr\delta\theta^2\le p_l<p_t+zr\delta\theta^2$, we have:

$$\widetilde{\pi}_E^{RL}(p_t,p_l)=(p_t-w)z(1-\delta)(1-p_t)+(p_l-w-r)z\delta\big(1-p_l+zr\delta\theta^2\big).\tag{A.9}$$

By checking that $\frac{\partial^2\widetilde{\pi}_E^{RL}(p_t,p_l)}{\partial p_l^2}=-2z\delta<0$, we obtain the e-tailer's optimal live-streaming price given the traditional retail price as $\widetilde{p}_l^{RL}(p_t)$. Solving the FOC, we obtain $p_l=\frac{1+w+r+zr\delta\theta^2}{2}$ and compare it with the critical values. Given that $r<\bar{r}$, we can obtain that the local maxima over $zr\delta\theta^2\le p_l<p_t+zr\delta\theta^2$ is:

$$\widetilde{p}_l^{RL}(p_t)=\begin{cases}p_t+zr\delta\theta^2,\ if\ p_t<\dfrac{1+w+r\big(1-z\delta\theta^2\big)}{2},\\ \dfrac{1+w+r+zr\delta\theta^2}{2},\ if\ p_t\ge\dfrac{1+w+r\big(1-z\delta\theta^2\big)}{2}.\end{cases}\tag{A.10}$$

Consider the first case in (A.8). If $p_l<zr\delta\theta^2$, we have:

$$\widetilde{\pi}_E^{RL}(p_t,p_l)=(p_t-w)z(1-\delta)(1-p_t)+(p_l-w-r)z\delta,$$

where $\frac{\partial\widetilde{\pi}_E^{RL}(p_t,p_l)}{\partial p_l}>0$, Thus, the local maxima over $p_l<zr\delta\theta^2$ is $zr\delta\theta^2$. Comparing the e-tailer's profits at the two local maxima, we can check that the e-tailer's optimal live-streaming price is attained at (A.10), and $\widetilde{e}^{RL}=zr\delta\theta$. Substitute (A.10) to (A.9). If $p_t<\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$, we have:

$\widetilde{\pi}_E^{RL}(p_t)=(1-p_t)z\big(p_t-w-r\delta\big(1-z\delta\theta^2\big)\big)$.

It is easy to check that $\widetilde{\pi}_E^{RL}(p_t)$ is concave. By solving the FOC, we obtain $p_t=\frac{1+w+r\delta\big(1-z\delta\theta^2\big)}{2}$, which can be larger than $\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$. Comparing the critical value, we have the local maxima over $p_t<\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$ given below:

$$\widetilde{p}_t^{RL}(w)=\begin{cases}\dfrac{1+w+r\delta\big(1-z\delta\theta^2\big)}{2},\ if\ \Gamma<\dfrac{1}{z},\\ \dfrac{1+w+r\big(1-z\delta\theta^2\big)}{2},\ if\ \Gamma\ge\dfrac{1}{z}.\end{cases}$$

If $p_t\ge\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$, we have:

$\widetilde{\pi}_E^{RL}(p_t)=(p_t-w)z(1-\delta)(1-p_t)+\frac{z\delta\big(1-w-r+zr\delta\theta^2\big)^2}{4}$.

It is easy to check that $\widetilde{\pi}_E^{RL}(p_t)$ is concave. By solving the FOC, we obtain $p_t=\frac{1+w}{2}$, which can be smaller than $\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$. Thus, the local maxima over $p_t\ge\frac{1+w+r\big(1-z\delta\theta^2\big)}{2}$ is shown as follows:

$$\widetilde{p}_t^{RL}(w)=\begin{cases}\dfrac{1+w+r\big(1-z\delta\theta^2\big)}{2},\ if\ \Gamma<\dfrac{1}{z},\\ \dfrac{1+w}{2},\ if\ \Gamma\ge\dfrac{1}{z}.\end{cases}$$

Comparing the e-tailer's profits at the two local maxima, we can check the optimal traditional price and substitute it to (A.10). Therefore, the e-tailer's optimal retail prices are as follows:

$$\big(\widetilde{p}_t^{RL}(w),\widetilde{p}_l^{RL}(w)\big)=\begin{cases}\left(\dfrac{1+w+r\delta\big(1-z\delta\theta^2\big)}{2},\dfrac{1+w+r\delta\big(1-z\delta\theta^2+2z\theta^2\big)}{2}\right),\ if\ \Gamma<\dfrac{1}{z},\\ \left(\dfrac{1+w}{2},\dfrac{1+r+w+zr\delta\theta^2}{2}\right),\ if\ \Gamma\ge\dfrac{1}{z}.\end{cases}$$

Consider the brand's optimal wholesale price. By putting the retail prices and the sales effort into (10), the brand's profit is given by:

$\widetilde{\pi}_B^{RL}(w)=\frac{1}{2}wz\big(1-w-r\delta\big(1-z\delta\theta^2\big)\big)$.

It is easy to check that $\frac{d^2\widetilde{\pi}_B^{RL}(w)}{dw^2}=-z<0$, and thus the optimal profit is attained at the FOC $\widetilde{w}^{RL}=\frac{1-r\delta\big(1-z\delta\theta^2\big)}{2}$. Putting $\widetilde{w}^{RL}$ into $\big(\widetilde{p}_t^{RL}(w),\widetilde{p}_l^{RL}(w)\big)$

derives Lemma 4. ▌

Proof of Lemma 5. We derive the brand's optimal retail prices from the optimal sales effort (A.8). Regarding the third case in (A.8), by putting $\widetilde{e}^{ML}(p_t,p_l)$ to (12), we have:

$$\widetilde{\pi}_B^{ML}(p_t,p_l) = (1-\lambda)p_t z(1-\delta)(1-p_t) + ((1-\lambda)p_l - r)z\delta(1-p_t),$$

where $\frac{\partial \widetilde{\pi}_B^{ML}(p_t,p_l)}{\partial p_l} > 0$. This implies the brand will choose a sufficiently high p_l, which leads to "zero profit" for the streamer and hence cannot be optimal. Thus, it suffices to consider the first two cases in (A.8). Consider the second case in (A.8). If $zr\delta\theta^2 \le p_l < p_t + zr\delta\theta^2$, we have:

$$\widetilde{\pi}_B^{ML}(p_t,p_l) = (1-\lambda)p_t z(1-\delta)(1-p_t) + ((1-\lambda)p_l - r)z\delta\left(1-p_l+zr\delta\theta^2\right). \tag{A.11}$$

By checking that $\frac{\partial^2 \widetilde{\pi}_B^{ML}(p_t,p_l)}{\partial p_l^2} = -2z\delta(1-\lambda) < 0$, we obtain the brand's optimal live-streaming price given the traditional retail price as $\widetilde{p}_l^{ML}(p_t)$. Solving the FOC, we obtain $p_l = \frac{r+\left(1+zr\delta\theta^2\right)(1-\lambda)}{2(1-\lambda)}$ and compare it with the critical values. Given that $r < \bar{r}$, the local maxima over $zr\delta\theta^2 \le p_l < p_t + zr\delta\theta^2$ is:

$$\widetilde{p}_l^{ML}(p_t) = \begin{cases} p_t + zr\delta\theta^2, \ if\ p_t < \dfrac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}, \\ \dfrac{r+\left(1+zr\delta\theta^2\right)(1-\lambda)}{2(1-\lambda)}, \ if\ p_t \ge \dfrac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}. \end{cases} \tag{A.12}$$

Consider the first case in (A.8). If $p_l < zr\delta\theta^2$, we have:

$$\widetilde{\pi}_B^{ML}(p_t,p_l) = (1-\lambda)p_t z(1-\delta)(1-p_t) + ((1-\lambda)p_l - r)z\delta,$$

where $\frac{\partial \widetilde{\pi}_B^{ML}(p_t,p_l)}{\partial p_l} > 0$, Thus, the local maxima over $p_l < zr\delta\theta^2$ is $zr\delta\theta^2$. Comparing the brand's profits at the two local maxima, we can check the brand's optimal live-streaming price is attained at (A.12), and $\widetilde{e}^{ML} = zr\delta\theta$. Substitute (A.12) to (A.11). If $p_t < \frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$, we have:

$$\widetilde{\pi}_B^{ML}(p_t,p_l) = (1-\lambda)p_t z(1-\delta)(1-p_t) + \left((1-\lambda)\left(p_t+zr\delta\theta^2\right) - r\right)z\delta(1-p_t).$$

It is easy to check that $\widetilde{\pi}_B^{ML}(p_t)$ is concave. By solving the FOC, we obtain $p_t = \frac{1-\lambda+r\delta\left(1-z\delta\theta^2(1-\lambda)\right)}{2(1-\lambda)}$, which can be larger than $\frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$. Comparing the critical value, we have the local maxima over $p_t < \frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$ is:

$$\widetilde{p}_t^{ML} = \begin{cases} \dfrac{1-\lambda+r\delta\left(1-z\delta\theta^2(1-\lambda)\right)}{2(1-\lambda)}, \ if\ \Gamma < \dfrac{1}{z(1-\lambda)}, \\ \dfrac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}, \ if\ \Gamma \ge \dfrac{1}{z(1-\lambda)}. \end{cases}$$

If $p_t \ge \frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$, we have:

$$\widetilde{\pi}_B^{ML}(p_t) = (1-\lambda)p_t z(1-\delta)(1-p_t) + \frac{z\delta\left(\left(1+zr\delta\theta^2\right)(1-\lambda)-r\right)^2}{4(1-\lambda)}.$$

It is easy to check that $\widetilde{\pi}_B^{ML}(p_t)$ is concave. By solving the FOC, we obtain $p_t = \frac{1}{2}$, which can be smaller than $\frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$. Comparing the critical value, we have the local maxima over $p_t \ge \frac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}$ is:

$$\widetilde{p}_t^{ML} = \begin{cases} \dfrac{(1-\lambda)\left(1-zr\delta\theta^2\right)+r}{2(1-\lambda)}, \ if\ \Gamma < \dfrac{1}{z(1-\lambda)}, \\ \dfrac{1}{2}, \ if\ \Gamma \ge \dfrac{1}{z(1-\lambda)}. \end{cases}$$

Comparing the local optimal value derives the brand's best responses as follows:

$$\widetilde{p}_t^{ML} = \begin{cases} \dfrac{1-\lambda+r\delta\left(1-z\delta\theta^2(1-\lambda)\right)}{2(1-\lambda)}, \ if\ \Gamma < \dfrac{1}{z(1-\lambda)}, \\ \dfrac{1}{2}, \ if\ \Gamma \ge \dfrac{1}{z(1-\lambda)}. \end{cases}$$

Putting $\widetilde{p}_t^{ML}$ into $\widetilde{p}_l^{ML}(p_t)$ derives Lemma 5. ▌

Proof of Corollary 3. Consider the reselling format. From Lemma 4, we establish the proof in two cases. If $\Gamma < 1/z$, we have $\widetilde{p}_l^{RL} - \widetilde{p}_t^{RL} = rz\delta\theta^2, \widetilde{p}_t^{RL} - \widetilde{p}_t^{RT} = \frac{r\delta\left(1-z\delta\theta^2\right)}{4}$, and $\widetilde{D}^{RL} - \widetilde{D}^{RT} = -\frac{zr\delta\left(1-z\delta\theta^2\right)}{4}$, which imply $\widetilde{p}_l^{RL} > \widetilde{p}_t^{RL} > \widetilde{p}_t^{RT}$ and $\widetilde{D}^{RL} < \widetilde{D}^{RT}$. If $\Gamma \ge 1/z$, we have $\widetilde{p}_l^{RL} - \widetilde{p}_t^{RL} = \frac{r\left(1+z\delta\theta^2\right)}{2}, \widetilde{p}_t^{RL} - \widetilde{p}_t^{RT} = -\frac{r\delta\left(1-z\delta\theta^2\right)}{4}$, and $\widetilde{D}^{RL} - \widetilde{D}^{RT} = -\frac{zr\delta\left(1-z\delta\theta^2\right)}{4}$, which imply $\widetilde{p}_l^{RL} > \widetilde{p}_t^{RL} \ge \widetilde{p}_t^{RT}$ and $\widetilde{D}^{RL} \ge \widetilde{D}^{RT}$.

Consider the marketplace format. From Lemma 5, we establish the proof in two cases. If $\Gamma < \frac{1}{z(1-\lambda)}$, we have $\widetilde{p}_l^{ML} - \widetilde{p}_t^{ML} = rz\delta\theta^2, \widetilde{p}_t^{ML} - \widetilde{p}_t^{MT} =$

$\frac{r\delta\left(1-(1-\lambda)z\delta\theta^2\right)}{2(1-\lambda)}$, and $\widetilde{D}^{ML}-\widetilde{D}^{MT}=-\frac{zr\delta\left(1-z\delta\theta^2(1-\lambda)\right)}{2(1-\lambda)}$, which imply $\widetilde{p}_l^{ML}>\widetilde{p}_t^{ML}>\widetilde{p}_t^{MT}$ and $\widetilde{D}^{ML}<\widetilde{D}^{MT}$. If $\Gamma\geq\frac{1}{z(1-\lambda)}$, we have $\widetilde{p}_l^{ML}-\widetilde{p}_t^{ML}=\frac{1}{2}r\left(z\delta\theta^2+\frac{1}{1-\lambda}\right)$, $\widetilde{p}_t^{ML}-\widetilde{p}_t^{MT}=0$, and $\widetilde{D}^{ML}-\widetilde{D}^{MT}=-\frac{zr\delta\left(1-z\delta\theta^2(1-\lambda)\right)}{2(1-\lambda)}$, which imply $\widetilde{p}_l^{ML}>\widetilde{p}_t^{ML}=\widetilde{p}_t^{MT}$ and $\widetilde{D}^{ML}\geq\widetilde{D}^{MT}$. ▌

Proof of Proposition 5. We first present the optimal profits for the supply chain members without supply risks in the following:

$$\widetilde{\Pi}_E^{RL}=\begin{cases}\dfrac{\left(1-r\delta\left(1-\delta\theta^2\right)\right)^2}{16},\ \textit{if}\ \Gamma<1,\\ \dfrac{1-2r\delta\left(1-\delta\theta^2\right)+\delta r^2(4-3\delta)\left(1-\delta\theta^2\right)^2}{16},\ \textit{if}\ \Gamma\geq1.\end{cases}$$

$$\widetilde{\Pi}_E^{ML}=\begin{cases}\dfrac{\lambda\left(\left(1-\lambda+r\delta^2\theta^2(1-\lambda)\right)^2-\delta^2r^2\right)}{4(1-\lambda)^2},\ \textit{if}\ \Gamma<\dfrac{1}{1-\lambda},\\ \dfrac{\lambda\left((1-\lambda)^2\left(1+r\delta^2\theta^2\left(2+r\delta\theta^2\right)\right)-r^2\delta\right)}{4(1-\lambda)^2},\ \textit{if}\ \Gamma\geq\dfrac{1}{1-\lambda}.\end{cases}$$

$$\widetilde{\Pi}_B^{ML}=\begin{cases}\dfrac{\left(1-\lambda-r\delta\left(1-\delta\theta^2(1-\lambda)\right)\right)^2}{4(1-\lambda)},\ \textit{if}\ \Gamma<\dfrac{1}{1-\lambda},\\ \dfrac{(1-\delta)(1-\lambda)}{4}+\dfrac{\delta\left(1-\lambda-r+r\delta\theta^2(1-\lambda)\right)^2}{4(1-\lambda)},\ \textit{if}\ \Gamma\geq\dfrac{1}{1-\lambda}.\end{cases}$$

We start with analyzing the optimal channel strategy. Consider the e-tailer's optimal profits. If $\Gamma<1$, we have $\widetilde{\Pi}_E^{RL}-\overline{\Pi}_E^{RT}=\frac{-r\delta\left(1-\delta\theta^2\right)\left(2-r\delta\left(1-\delta\theta^2\right)\right)}{16}<0$. If $\Gamma\geq1$, we have $\widetilde{\Pi}_E^{RL}-\overline{\Pi}_E^{RT}=\frac{r\delta\left(\delta\theta^2-1\right)\left(2+r(4-3\delta)\left(\delta\theta^2-1\right)\right)}{16}\geq0$. Consider the brand's optimal profits. If $\Gamma<\frac{1}{1-\lambda}$, we have $\widetilde{\Pi}_B^{ML}-\overline{\Pi}_B^{MT}=\frac{-r\delta\left(1-\delta\theta^2(1-\lambda)\right)\left(2(1-\lambda)-r\delta\left(1-\delta\theta^2(1-\lambda)\right)\right)}{4(1-\lambda)}$. From $\lambda\leq1-2r$, it is easy to check that $\widetilde{\Pi}_B^{ML}<\overline{\Pi}_B^{MT}$. If $\Gamma\geq\frac{1}{1-\lambda}$, we have $\widetilde{\Pi}_B^{ML}-\overline{\Pi}_B^{MT}=\frac{\delta\left(2(1-\lambda)+r\left(\delta\theta^2(1-\lambda)-1\right)\right)\left(r\left(\delta\theta^2(1-\lambda)-1\right)\right)}{4(1-\lambda)}\geq0$.

Next, we derive the e-tailer's optimal model strategy by considering three cases.

Case (i) $\Gamma<1$. In this case, neither the e-tailer nor the brand introduces a live-streaming channel, and $\overline{\Pi}_E^{RT}-\overline{\Pi}_E^{MT}=\frac{1-4\lambda}{16}$. Thus, we have $\overline{\Pi}_E^{RT}-\overline{\Pi}_E^{MT}\geq0$ if $\lambda\leq\frac{1}{4}$, and $\overline{\Pi}_E^{RT}-\overline{\Pi}_E^{MT}<0$ if $\lambda>\frac{1}{4}$.

Case (ii) $1\leq\Gamma<\frac{1}{1-\lambda}$. In this case, the e-tailer introduces a live-streaming channel but the brand does not, and we have:

$$\widetilde{\Pi}_E^{RL}-\overline{\Pi}_E^{MT}=\frac{\delta r^2(4-3\delta)\left(\delta\theta^2\right)^2+2\delta r(1-r(4-3\delta))\delta\theta^2+1-\delta r(2-r(4-3\delta))-4\lambda}{16},$$

the sign of which depends on the numerator. Let $\widetilde{j}(x)=\delta r^2(4-3\delta)x^2+2\delta r(1-r(4-3\delta))x+1-\delta r(2-r(4-3\delta))-4\lambda$, which increases in $\left[1,\frac{1}{1-\lambda}\right)$, and satisfies $\widetilde{j}(1)=1-4\lambda$, and $\widetilde{j}\left(\frac{1}{1-\lambda}\right)=\frac{r^2\delta(4-3\delta)+2r\delta(1-r(4-3\delta))(1-\lambda)+(1-\lambda)^2(1-r\delta(2-r(4-3\delta))-4\lambda)}{(1-\lambda)^2}$. We then evaluate the sign of $\widetilde{j}(x)$ according to the value of λ. If $\lambda\leq\frac{1}{4}$, we can check that $\widetilde{j}(1)\geq0$ and $\widetilde{j}\left(\frac{1}{1-\lambda}\right)>0$. Then, we have $\widetilde{\Pi}_E^{RL}-\overline{\Pi}_E^{MT}\geq0$. If $\lambda>\frac{1}{4}$, we can check that $\widetilde{j}(1)<0$, and $\widetilde{j}\left(\frac{1}{1-\lambda}\right)$ can be either greater or smaller than 0. Let $\widetilde{\Gamma}_1^{'}$ be the unique solution to $\widetilde{j}(x)=0$ and let $\widetilde{\Gamma}_1=\min\left\{\widetilde{\Gamma}_1^{'},\frac{1}{1-\lambda}\right\}$. Then, we have $\widetilde{\Pi}_E^{RL}-\overline{\Pi}_E^{MT}<0$ if $\lambda>\frac{1}{4}$ and $1\leq\Gamma<\widetilde{\Gamma}_1$.

Case (iii) $\Gamma\geq\frac{1}{1-\lambda}$. In this case, both the e-tailer and brand introduce a live-streaming channel, and we have:

$$\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}=\frac{\widetilde{A}\left(\delta\theta^2\right)^2+\widetilde{B}\delta\theta^2+\widetilde{C}}{16(1-\lambda)^2},$$

where $\widetilde{A}=r^2\delta(4-3\delta-4\lambda)(1-\lambda)^2$, $\widetilde{B}=2r\delta(1-r(4-3\delta)-4\lambda)(1-\lambda)^2$, $\widetilde{C}=(1-\lambda)^2(1-r\delta(2-r(4-3\delta))-4\lambda)+4r^2\delta\lambda$, and the denominator is positive. Let $\widetilde{k}(x)=\widetilde{A}x^2+\widetilde{B}x+\widetilde{C}$, and the upper bound $\overline{\Gamma}=\min\left\{\frac{1}{r},\frac{1-2r+r\delta}{r\delta}\right\}$. We discuss the sign of $\widetilde{k}(x)$ according to the value of λ. (a) If $\lambda<\frac{4-3\delta}{4}$ then we can check $\widetilde{A}>0$ and $\widetilde{k}(x)$ is convex. If $\widetilde{k}(x)$ has no root or only one root, then $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}\geq0$, and in this case, we let $\widetilde{\Gamma}_2=\frac{1}{1-\lambda}$. Otherwise, we let $\widetilde{\Gamma}^{'}$ and $\widetilde{\Gamma}^{''}$, where $\widetilde{\Gamma}^{'}<\widetilde{\Gamma}^{''}$, be the two roots to $\widetilde{k}(x)=0$. We can check that $\widetilde{\Gamma}^{'}<\frac{1}{1-\lambda}$. We then specify the value of $\widetilde{\Gamma}_2$ for different cases of zero roots. If $\widetilde{\Gamma}^{''}\leq\frac{1}{1-\lambda}$, then $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}\geq0$ over $\Gamma\in\left[\frac{1}{1-\lambda},\overline{\Gamma}\right)$, and we let $\widetilde{\Gamma}_2=\frac{1}{1-\lambda}$. If $\frac{1}{1-\lambda}<\widetilde{\Gamma}^{''}<\overline{\Gamma}$, then $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}<0$ if and only if $\Gamma<\widetilde{\Gamma}^{''}$, and we let $\widetilde{\Gamma}_2=\widetilde{\Gamma}^{''}$. If $\widetilde{\Gamma}^{''}\geq\overline{\Gamma}$, then $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}<0$ over $\Gamma\in\left[\frac{1}{1-\lambda},\overline{\Gamma}\right)$, and we let $\widetilde{\Gamma}_2=\overline{\Gamma}$. Summarizing the subcases above, we have $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}<0$ if and only if $\Gamma<\widetilde{\Gamma}_2$. (b) if $\lambda\geq\frac{4-3\delta}{4}$, we can check that $\widetilde{\Pi}_E^{RL}-\widetilde{\Pi}_E^{ML}<0$ over $\Gamma\in\left[\frac{1}{1-\lambda},\overline{\Gamma}\right)$, and we let $\widetilde{\Gamma}_2=\overline{\Gamma}$. Summarizing the three subcases in case (iii), we identify the threshold

$\widetilde{\Gamma}_2$ such that $\overline{\Pi}_E^{RL} < \overline{\Pi}_E^{ML}$ if and only if $\frac{1}{1-\lambda} \le \Gamma < \widetilde{\Gamma}_2$.

To summarize, Proposition 5 is proven. ▌

Proof of Lemma 6. Following the same procedure in the main model, we first analyze the optimal sales effort, retail prices, and wholesale price. These optimal solutions are derived through backwards induction and are summarized in Table A.3.

Table A.3
The equilibrium solutions with supply viability improvement.

	$\widehat{e}^{XL}$	$\widehat{p}_t^{XL}$	$\widehat{p}_l^{XL}$	$\widehat{w}^{XL}$
Reselling format	$\frac{\alpha\delta\theta}{4(1-\alpha\delta\theta^2)}$	$\frac{3}{4}$	$\frac{3-2\alpha\delta\theta^2}{4(1-\alpha\delta\theta^2)}$	$\frac{1}{2}$
Marketplace format	$\frac{\alpha\delta\theta(1-\lambda)}{2(1-(1-\lambda)\alpha\delta\theta^2)}$	$\frac{1}{2}$	$\frac{1}{2(1-(1-\lambda)\alpha\delta\theta^2)}$	–

By putting the results in Table A.3 into (15) and (17), we obtain

$$\widehat{\pi}_E^{RL}(\beta) = \frac{\beta\left(1-\alpha\delta\left(1+(1-\delta)\theta^2\right)\right)}{16\left(1-\alpha\delta\theta^2\right)} - \frac{1}{2}\beta^2,$$

$$\widehat{\pi}_B^{ML}(\beta) = \frac{\beta(1-\lambda)\left(1-\alpha\left(\delta+(1-\lambda)(1-\delta)\delta\theta^2\right)\right)}{4\left(1-(1-\lambda)\alpha\delta\theta^2\right)} - \frac{1}{2}\beta^2.$$

We can check that $\frac{d^2\widehat{\pi}_E^{RL}(\beta)}{d\beta^2} = -1 < 0$ and $\frac{d^2\widehat{\pi}_B^{ML}(\beta)}{d\beta^2} = -1 < 0$, which implies the concavity of $\widehat{\pi}_E^{RL}(\beta)$ and $\widehat{\pi}_B^{ML}(\beta)$. By solving the first-order condition (FOC), we obtain the optimal viability levels as stated in Lemma 6. ▌

Proof of Corollary 4. From Lemma 6, we have (i) $\frac{\partial\widehat{\beta}^{RL}}{\partial\alpha} = \frac{-\delta\left(1-\delta\theta^2\right)}{16\left(1-\alpha\delta\theta_2^2\right)^2}$ and $\frac{\partial\widehat{\beta}^{ML}}{\partial\alpha} = \frac{-\delta(1-\lambda)\left(1-(1-\lambda)\delta\theta^2\right)}{4\left(1-(1-\lambda)\alpha\delta\theta^2\right)^2}$. Let $\widehat{\Gamma}_1^R = 1$, $\widehat{\Gamma}_1^M = \frac{1}{1-\lambda}$. Then, we have $\frac{\partial\widehat{\beta}^{RL}}{\partial\alpha} < 0$ if and only if $\Gamma < \widehat{\Gamma}_1^R$, $\frac{\partial\widehat{\beta}^{ML}}{\partial\alpha} < 0$ if and only if $\Gamma < \widehat{\Gamma}_1^M$; (ii) $\frac{\partial\widehat{\beta}^{RL}}{\partial\delta} = \frac{-\alpha^2\left(\delta\theta^2-\frac{1}{\alpha}\right)+1-\alpha}{16\left(1-\alpha\delta\theta^2\right)^2}$ and $\frac{\partial\widehat{\beta}^{ML}}{\partial\delta} = \frac{-\alpha^2(1-\lambda)^3\left(\delta\theta^2-\frac{1}{\alpha(1-\lambda)}\right)+(1-\alpha)(1-\lambda)}{4\left(1-\alpha\delta\theta^2(1-\lambda)\right)^2}$. Let $\widehat{\Gamma}_2^R = \frac{1-\sqrt{1-\alpha}}{\alpha}$, $\widehat{\Gamma}_2^M = \frac{1-\sqrt{1-\alpha}}{\alpha(1-\lambda)}$. Thus, we have $\frac{\partial\widehat{\beta}^{RL}}{\partial\delta} < 0$ if and only if $\Gamma < \widehat{\Gamma}_2^R$, $\frac{\partial\widehat{\beta}^{ML}}{\partial\delta} < 0$ if and only if $\Gamma < \widehat{\Gamma}_2^M$. (iii) It is easy to check that $\frac{\partial\widehat{\beta}^{RL}}{\partial\theta} = \frac{(1-\alpha)\alpha\delta^2\theta}{8\left(1-\alpha\delta\theta^2\right)^2} > 0$, $\frac{\partial\widehat{\beta}^{ML}}{\partial\theta} = \frac{(1-\alpha)(1-\lambda)^2\alpha\delta^2\theta}{2\left(1-\alpha\delta\theta^2(1-\lambda)\right)^2} > 0$. To summarize, Corollary 4 is proven. ▌

Proof of Corollary 5. We first list the equilibrium solutions without introducing a live-streaming channel.

Table A.4
The optimal profits with and without live-streaming channel in context of supply viability improvement.

	$\widehat{\pi}_E^{RT}$	$\widehat{\pi}_B^{MT}$	$\widehat{\pi}_E^{RL}$	$\widehat{\pi}_B^{ML}$
Reselling format	$\frac{1}{512}$	$\frac{1}{128}$	$\frac{\left(1-\alpha\delta\left(1+(1-\delta)\theta^2\right)\right)^2}{512\left(1-\alpha\delta\theta^2\right)^2}$	$\frac{1-\alpha\delta\left(1+(1-\delta)\theta^2\right)}{128\left(1-\alpha\delta\theta^2\right)}$
Marketplace format	$\frac{\lambda(1-\lambda)}{16}$	$\frac{(1-\lambda)^2}{32}$	$\frac{\lambda(1-\lambda)\left(1-(1-\lambda)(1-\delta)\alpha\delta\theta^2-\alpha\delta\right)\left(1-(1-\lambda)(1-\delta)\alpha\delta\theta^2\right)}{16\left(1-(1-\lambda)\alpha\delta\theta^2\right)^2}$	$\frac{(1-\lambda)^2\left(1-\alpha\delta\left(1+(1-\lambda)(1-\delta)\theta^2\right)\right)^2}{32\left(1-(1-\lambda)\alpha\delta\theta^2\right)^2}$

From Table A.4, we calculate the difference in profits with and without live-streaming channel under the reselling format, we have $\widehat{\pi}_E^{RL} - \widehat{\pi}_E^{RT} = \frac{\alpha\delta\left(\delta\theta^2-1\right)\left(2-\alpha\delta\left(1+(2-\delta)\theta^2\right)\right)}{512\left(1-\alpha\delta\theta^2\right)^2}$, the sign of which depends on "$\delta\theta^2 - 1$". We have $\widehat{\pi}_E^{RL} \ge \widehat{\pi}_E^{RT}$ if and only if $\Gamma \ge \widehat{\Gamma}_1^R$.

Similarly, under the marketplace format, we have $\widehat{\pi}_B^{ML} - \widehat{\pi}_B^{MT} = \frac{\alpha\delta\left((1-\lambda)\delta\theta^2-1\right)\left(2-\alpha\delta\left(1+(2-\delta)(1-\lambda)\theta^2\right)\right)(1-\lambda)^2}{32\left(1-(1-\lambda)\alpha\delta\theta^2\right)^2}$, the sign of which depends on "$(1-\lambda)\delta\theta^2 - 1$". We have $\widehat{\pi}_B^{ML} \ge \widehat{\pi}_B^{MT}$ if and only if $\Gamma \ge \widehat{\Gamma}_1^M$. ▌

Data availability

No data was used for the research described in the article.

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